Asymmetric Impact of Matching Technology on Influencer Marketing: Implications for Platform Revenue *

Jessie Liu†1 and Yi Liu‡2

[1]Johns Hopkins University

[2]University of Wisconsin–Madison

Abstract

This paper explores the impact of using advanced technology such as artificial intelligence (AI) to match marketers with social media influencers. We develop a theoretical model to examine how matching accuracy affects the competition between influencers and the profitability of a social media platform. Our findings show that improving matching accuracy may not always benefit the platform, especially for platforms with intermediate follower density. Two opposing effects of technology improvement affect the prices of influencer marketing campaigns: advanced technology such as AI enhances the matching between influencers and marketers, but also intensifies competition between different types of influencers. The overall effect on prices can be negative for some influencers due to the *asymmetric* nature of such matching technology: the matching outcome for influencers with a narrower audience ("niche" influencers) is more sensitive to matching accuracy than that for those with a broader audience ("general" influencers). As a result, more niche influencers begin to participate in marketing campaigns when matching accuracy improves, which reduces the prices offered by sufficiently general influencers and may lead to a decline in platform revenue. Additionally, we found that adjusting commission rates in response to technology improvements could help mitigate the negative impact, although it may not eliminate it entirely. Our findings offer valuable insights for social media platforms seeking to remain competitive in the influencer marketing landscape.

Keywords: matching, influencer marketing, competition, technology strategy

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†Email: jzliu@jhu.edu

‡Email: yliu2396@wisc.edu

1 Introduction

Social media usage was up by almost 60% since the pandemic lockdown (Nelson Media Report, 2020). With that, influencer marketing—where marketers collaborate with Internet "influencers" on social media to promote their products—has now become one of the top trends in online advertising. Such influencers have a significant number of followers consuming the content they produce on social media platforms, making them an ideal conduit to potential customers. When collaborating with marketing firms, the influencers create specific "content" (such as videos, blogs, etc.) to reach targeted customers, drive engagement, and ultimately convert sales. In 2022, the global market size of influencer marketing was valued at a record of $16.4 billion (Influencer Marketing Hub, 2022). Social media platforms also gain from influencer marketing as it provides a new revenue source through service fees (WebFX, 2023; Southern Metropolis Daily, 2020).

Since influencers have widely heterogeneous characteristics, the effectiveness of a marketer's influencer marketing campaign depends critically on the marketer's choice of the "right" influencer. Avery and Israeli (2020, p. 7) point out in a Harvard Business School technical note on influencer marketing that "according to marketers, one of the most frustrating challenges associated with influencer marketing is choosing the right influencers with whom to work." The difficulty of finding the "right" influencer lies in the fact that there are a large and increasing number of influencers, especially micro-influencers, available on the Internet. For example, by October 2020, there were 500,000 active influencers on Instagram[1], and in 2017, there were 6.4 million Twitter users who had 1K-10K followers, which comprise 2% of the total user base on Twitter.[2] One proposed solution to improving the matching between marketers and influencers is for platforms to offer advanced technology such as artificial intelligence (AI) or machine learning algorithms (Farseev et al., 2018; Sweet et al., 2019). Social media platforms care about the match between brands and influencers because they collect a percentage service fee from sales induced by on-the-platform influencer marketing (Southern Metropolis Daily, 2020). As Krishna Subramanian, the co-founder and CEO of an end-to-end influencer marketing platform points out, "[t]he big trend we see is that brands want easier ways to find creators through leveraging data." However, other executives have expressed concerns about adopting AI in influencer marketing, since it may generate inaccurate predictions and campaign errors (Barkho, 2023). With that being said, would improving AI's prediction accuracy fully resolve practitioners' concerns? In this paper, we investigate whether a social media

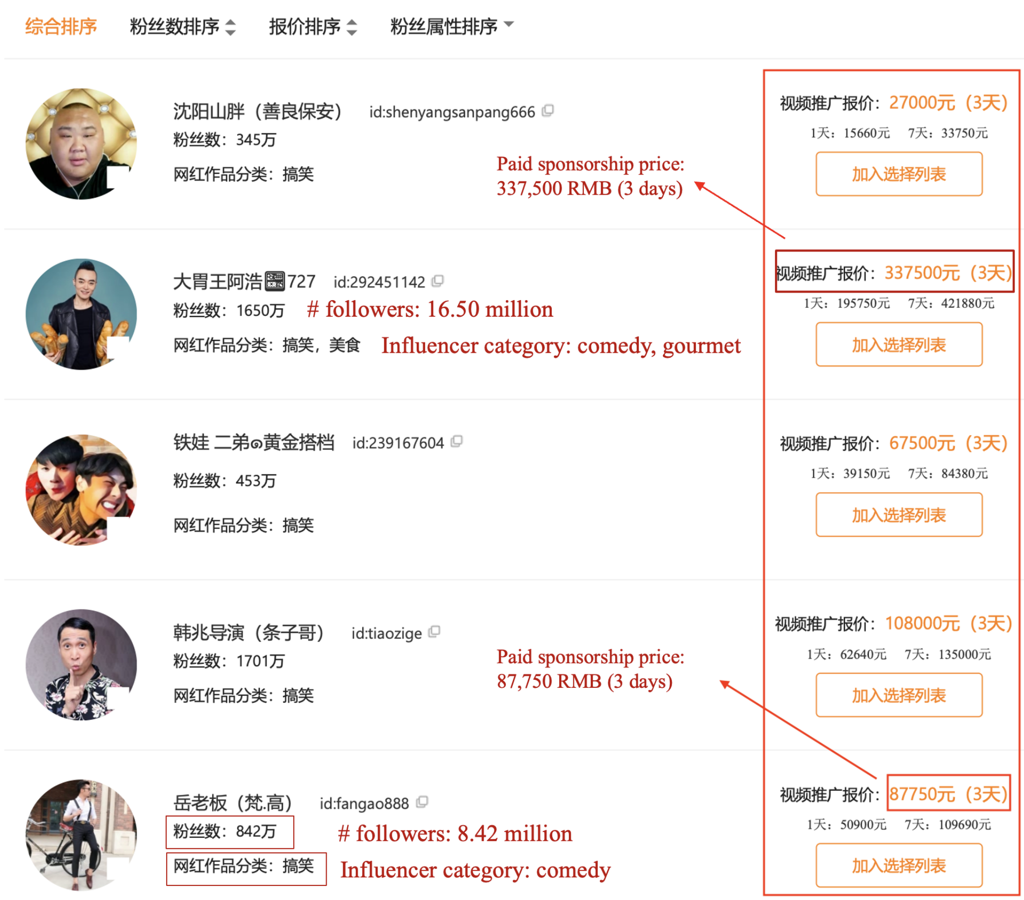
[1] `https://www.omnicoreagency.com/instagram-statistics/`.
[2] `https://askwonder.com/research/social-influencers-10k-100k-followers-world-us-jzrt5l5im#`.

platform should embrace the advancement in matching technology[3] that helps marketers find the "right" influencers. Specifically, does a more accurate matching technology always bring in a higher revenue for the platform? Moreover, how does such technology improvement influence the platform's other strategic decisions such as setting the commission fee for influencer marketing campaigns?

To address these questions, we develop a theoretical model that captures four salient features of influencer marketing. First, each influencer on the social media platform has a follower base with a spectrum of traits and interests. Influencers are heterogeneous in their content broadness, or how dispersed their followers' interests are. Some influencers create content focused on "a cluster of subjects" inside a specific domain. Canadian influencer Amannda Julia, for example, managed an Instagram channel that is mainly about horses (Talita, 2018). Others "transit between different topics and off-topics" (Talita, 2018). A lifestyle influencer Laura Hunt, for example, described her Instagram posts as "pieces of my life displayed in grids" (Izea, 2020). In our model, this parameter of content broadness characterizes a spectrum of influencers from the former "niche" type to the latter "general" type. Meanwhile, marketers promoting products in a specific category have target consumers with distinct interests, from whom they expect to obtain the highest return on investment. For instance, the target consumers of a ski equipment brand are ski enthusiasts who enjoy winter sports. Although a general lifestyle influencer may have a subset of followers that aligns with the brand's target consumer base, a niche influencer focused on winter sports boasts a significantly more specialized audience, their interests closely mirroring those of the target consumers. Second, the platform's matching technology (AI) recommends a set of influencers to marketers based on the match of interests between influencers' followers and a marketer's target consumers (Barkho, 2023). Marketers choose among the recommended influencers and pay them a price proposed by the influencers in a competitive market for product endorsement. For example, Douyin, the Chinese version of TikTok under the same parent company, launched an AI-enabled influencer-marketer matching system called "Xingtu" in September 2018 (Chinese Social Media, 2020). Marketers specify their target consumers in the system; based on this information, the system recommends a list of influencers along with their proposed prices, as displayed in Figure 1. Third, influencers connect marketers and followers through product endorsement. When participating in the marketing campaigns, influencers engage their followers through videos or blogs and offer direct shop-from-content capabilities. Consumers can purchase the featured product without having to

[3]Throughout the paper, we refer to the matching technology capable of shifting the influencer-marketer match value as "AI" for convenience, similar to that in Choi et al. (2023).

Figure 1: Douyin's AI-recommended Influencers with Prices for Paid Sponsorship



leave the platform, making the content "seamlessly shoppable" (Avery and Israeli, 2020). We model this by assuming that marketers' revenue from the influencer marketing campaigns depends on the expected converted revenue of potential customers. Fourth, the platform charges a percentage commission fee to all participating influencers and marketers for providing the marketplace for those campaigns. For example, YouTube BrandConnect charges a 10% commission fee once an influencer is matched with a marketer for a marketing campaign (WebFX, 2023). Douyin and its major competitor Kuaishou charge a commission of 2-5% from influencers' on-the-platform campaigns in the form of "technical service fees" (Southern Metropolis Daily, 2020).

Our model makes the following predictions: First, the platform's revenue may decline as the matching accuracy improves, even when the cost of AI development is negligible. This is because improving AI enhances marketers' match value with influencers of different content broadness differently. Such an *asymmetric* impact of AI on influencers affects the distribution of equilibrium prices. On the one hand, a more accurate AI leads to better matches between the influencers and marketers as the interests of followers are closer to that of the target consumers. Better matches generate more sales for marketers, thereby not only motivating more marketers to buy endorsements from influencers, but also allowing influencers to charge a higher price. We refer to these two positive effects as the "market expansion effect" and the "matching enhancement effect," respectively. On the other hand, however, AI's matching enhancement effect can be asymmetric on influencers with different content broadness, as the expected revenue from niche influencers' marketing campaigns is more sensitive to the matching accuracy than general ones. Specifically, when more niche influencers become attractive to marketers due to the asymmetric enhancement effect, some general influencers have to lower their prices to compete with these new entrants for marketers' attention. We refer to this negative effect on prices as the "competition effect." The trade-off between the above three effects determines how a better AI affects the platform's total revenue. Therefore, we could observe a non-linear relationship between the platform's revenue and its matching accuracy. In particular, the density of followers on a platform moderates the trade-off between these effects, and thus serves as an important parameter a platform should watch when adopting advanced matching technology. Second, we also investigate how a platform's technology accuracy will interact with its choice of commission fee for influencer marketing. Intriguingly, the platform may increase (lower) its commission rate even when its revenue declines (increases) as AI becomes more accurate. We find that optimally adjusting the commission rate could mitigate, although not fully eliminate in some cases, the negative impact of AI improvement. Specifically, the platform may still suffer from a better brand-influencer AI, especially when there is heterogeneity in the influencers' opportunity costs incurred by those campaigns.

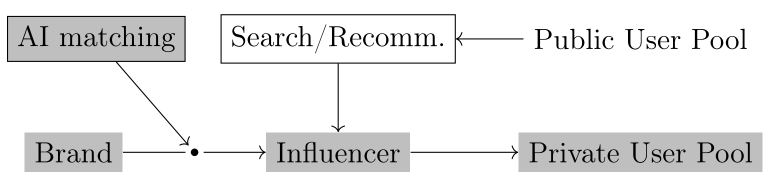
Our study provides some key managerial implications to social media platforms who aim to keep up with the industry trend of influencer marketing. First, although AI can facilitate matching influencers to marketers, developing such AI could sometimes backfire. The platform should proactively adjust its commission fee in order to take full advantage of or mitigate the damage from adopting an advanced matching technology. In addition, some platforms may be more willing to adopt such AI because of their unique influencer-follower composition, where the density of followers remains above a certain threshold. This generates a testable hypothesis for the timing of AI adoption across different social media platforms. In an era of growing consumer privacy concerns, influencer marketing enables targeting without the need to collect private data. This targeting value of influencers makes "influencer placements especially valuable in cases where consumers or brands are sensitive to privacy violations " (Campbell and Farrell, 2020, p. 473). Our study thus offers a timely discussion on how the advancement of new technology alters the incentives of marketers and social media platforms in adopting this new advertising model.

We contribute to four strands of literature. First, our paper contributes to the emerging research inspired by the growing interest in influencer marketing. For example, Yang et al. (2021) develops an algorithm to predict the effect of influencer video advertising on product sales. Nistor and Selove (2022) study how the information value of followers' comments affect an influencer's optimal endorsement policy. Fainmesser and Galeotti (2021) focus on how influencers trade-off endorsing paid products with creating organic content, which explains the phenomenon of the "rise of micro-influencers." Pei and Mayzlin (2022) study the optimal affiliation between a firm and influencers. Katona (2020) examines how firms compete for influencers in a network considering the overlap in the influencers' reach. Our study complements the above papers by investigating how the matching technology that reduces search frictions between marketers and influencers impacts the practice of influencer marketing.

Second, our paper contributes to the research on the business implications of applying new, advanced prediction technology such as AI to marketing contexts. For example, Lambrecht and Tucker (2019) empirically show that using an effectiveness-maximizing AI-algorithm in ad delivery may lead to gender-based discrimination. Lee and Hosanagar (2019) argue that using collaborative filter-based recommenders may induce a decrease in market diversity. Miklós-Thal and Tucker (2019) show that a more accurate demand prediction may not always facilitate collusion between sellers and thus may decrease or increase consumer welfare. Liu et al. (2023) discuss the unexpected consequences of applying advanced matching technology on a gig economy platform due to its information revelation effect. The above-mentioned papers show that the social and economic impact of prediction technology improvement is not as straightforward as it seems. Our study contributes to this stream of literature by exploring the nuanced impact of technology improvement on firms' profitability in the context of influencer marketing.

Third, beyond our contribution to research in influencer marketing and implications of prediction technology, we also contribute to a growing literature at the intersection of computer science, economics, and management science: search frictions in online markets. Search frictions in social media lead many platforms to develop curation algorithms to help users search for the most relevant content (in our case, influencers). Existing works have largely examined how to better design such algorithms matching users with influencers (Shardanand and Maes, 1995), or their effect on the content produced (see Latzer et al. (2016) for a survey). Fainmesser and Galeotti (2021) also discusses the role of search technology, albeit the one matching followers to influencers, in shaping the competition among influencers and their welfare. However, search frictions between brands and influencers are less studied, especially from a managerial perspective. Recent studies, mostly in computer science, begin to develop machine learning techniques for matching brands with social media influencers (Sweet et al., 2019). To the best of our knowledge, our paper is the first one that studies the economic implications of improving the accuracy of such algorithms. On a social media platform with both influencers and users, an influencer can deliver content to users in two different pools, which is illustrated in Figure 2. The first one is called the "private user pool," which consists of users who are followers of the influencer. For example, if a user subscribes to a YouTube channel, she is considered part of the "private user pool" of the YouTuber (influencer). In this pool, influencers "directly and frequently" interact with users because followers can consistently watch content generated by the same influencer.[4] The second one is called the "public user pool." This pool includes other users, i.e., non-followers of the influencer, who access the influencer's content via a search engine or algorithmic recommendations. Influencers only have limited interactions with users in this pool. In fact, delivering content to users in the "public user pool" is not a unique

Figure 2: Influencer Marketing



feature of influencer marketing, since this procedure is intrinsically similar to many other practices in digital marketing, such as Amazon's product recommendation engine. In other words, existing studies mostly focus on the search technology for the matching between influencers and users in the "public user pool." However, engaging users in the "private user pool" is exclusive to the practice of influencer marketing. This is the focus of our paper where the matching between brands and influencers plays a more important role.

Lastly, our paper is broadly related to the literature on targeted advertising. For instance, Shaffer and Zhang (1995) and Brahim et al. (2011) show in a horizontally differentiated context that targeted advertising (or coupon targeting) could lower firms' equilibrium profits compared to random advertising (or random distribution of coupons). In particular, a stream of research

[4] https://www.campaignasia.com/article/move-over-kols-kocs-and-private-domain-traffic-are-hot-in-china/454639.

has focused on the competitive implications of improved targeting accuracy through contextual advertising (Zhang and Katona, 2012; Chen and He, 2011; Sayedi et al., 2014). For example, Zhang and Katona (2012) show that heterogeneity in consumers' content preferences can be leveraged to reduce product market competition, especially when competition is intense. Similar to this stream of research, we explicitly model consumers' heterogeneous content preferences and study content relevance as a key parameter. However, our paper differs from the existing research in three distinct ways. First, contextual advertising helps marketers reach individual consumers with strong preferences for their product. By contrast, "firms" (marketers) in our paper cannot directly reach *individual* "consumers" (followers), but they advertise indirectly through influencers who provide unparalleled reach to a large number of followers. This modeling feature nicely reflects the fact that influencers can effectively communicate persuasive messages about the endorsed products to their followers. Matching technology improves the fit between influencers and marketers, but it does not change the composition of influencers' followers. Second, influencers in our framework operate as independent publishers on a platform (Zhang and Katona, 2012) and thus influencers, not marketers or the platform, set competitive prices for influencer marketing campaigns in our model, which is different from the existing literature on contextual advertising. Therefore, we focus on competition in the endorsement market, instead of the product market. Third, we propose a new mechanism where improved targeting technology in contextual advertising could change the competitive landscape in this market. Our key result on price competition between influencers arises from the asymmetric nature of how matching technology benefits different types of influencers, not from their ability to price discriminate, as discussed in the existing literature (Shaffer and Zhang, 1995; Brahim et al., 2011). To the best of our knowledge, we are the first to characterize this important feature of influencer marketing that gives rise to intensified competition through micro-founding the process of matching marketers with influencers based on their target audience.

The rest of the paper is organized as follows. Section 2 describes the model. We provide our main analysis in Section 3. In Section 4, we make several extensions to our benchmark model to generate additional insights. Finally, we conclude in Section 5.

2 Model

Consider a market consisting of a social media platform hosting influencers and their followers, as well as marketers that pay influencers to endorse their products. Followers have heterogeneous interests in products and are uniformly distributed on a unit circle. Influencers connect marketers and followers through product recommendations. Marketers search for influencers to buy their endorsement and the matching is mediated by the platform's AI recommender system. In what follows, we provide a model that links all these elements.

2.1 Influencers

Each consumer on the platform follows at least one influencer. Following Salop (1979), we denote an arbitrary consumer's location, i.e., the location of her interest in a product on the unit circle, as 0. Each consumer $x \in [0, 1)$ can be identified by her distance from 0 if she travels clockwise on the circle. For every x on the unit circle, there is a continuum of influencers with a follower base centered around x. That is, the followers of these influencers are located between $[x - \frac{n}{2}, x + \frac{n}{2}]$ with a density $\rho > 0$, where n is uniformly distributed on $[0, \frac{1}{2}]$[5] and it measures the broadness of interests among their followers or the content broadness of influencers. In other words, influencers are heterogeneous in the broadness of interests among their followers: some influencers (with small n) tend to gain their following in a specific niche by creating content relevant to that niche, whereas others (with large n) are more versatile and do not focus on one topic thereby attracting a diverse follower base. Hereafter, we refer to influencers with smaller n as more niche influencers and those with larger n as more general ones.[6] As such, each influencer n with followers centered around x has in total ρn followers.[7] A primary function of influencers is to create content with product endorsement. The strategy of each influencer n is to propose a price $p(n)$ for her sponsored content to marketers in a competitive market. This modeling choice is consistent with the fact that most influencers are paid with a flat fee for creating content or with in-kind services such as a free product or service (Activate, 2018).[8] If a marketer purchases an endorsement from an influencer, i.e., the marketer and the influencer "participate" in influencer marketing campaigns, the influencer creates the content and posts it. The influencer and the platform then share the proposed price based on a commission

[5] We restrict n to be less than $\frac{1}{2}$ to simplify our analysis. This assumption also eliminates the unrealistic cases where a less precise AI recommender system may improve the matching between an influencer's followers and a marketer's target group, which will be illustrated in Figure 3.

[6] For a description of niche vs. general influencers, see `https://medium.com/@talitarhein/niche-influencers-vs-general-influencers-7373cfa9f7c2`. Note that, in our baseline model, the parameter n effectively captures both the size and content broadness of an influencer. We will separately identify the role of heterogeneity in these two attributes of influencers in Section 4.5.

[7] We use n to refer to an influencer because an influencer's decision is independent of their followers' location x, which will become clear in Section 2.3.

[8] Regardless, in the Appendix on page A3, we show that our qualitative results remain unchanged when we allow for influencers to take a commission proportional to their sales.

rate. Let $\mathcal{N}$ denote the set of influencers who participate in the influencer marketing campaigns. If the influencer does not participate ($n \notin \mathcal{N}$), i.e., no marketers purchase her endorsement, she obtains a reservation utility denoted by γ_0 from her outside option. As such, influencer n's utility is given by

$$V(p(n)) = \begin{cases} (1-\delta)\, p(n) & \text{if } n \in \mathcal{N}, \\ \gamma_0 & \text{if } n \notin \mathcal{N}, \end{cases} \tag{1}$$

where $\delta \in (0,1)$ denotes the share of commission paid to the platform. To quickly deliver our main results, we let $\gamma_0 = 0$ in the baseline model. In Section 4, we discuss the robustness of our results when $\gamma_0 > 0$, and we further allow the influencer's opportunity cost to vary with the size of their followers and discuss its implications.

2.2 Marketers

In a representative product category, there is a continuum of mass C of marketers (or brands) whose target consumer is, without loss of generality, located at 0. Each marketer looks for up to one influencer on the platform to endorse their products. The return-on-investment of influencer marketing depends on two key factors: the cost of the program and the impact of the program on consumers' purchasing behaviors (Avery and Israeli, 2020). Marketers are heterogeneous in the fixed cost, denoted by c, that they need to incur in order to launch such campaigns. This cost includes compliance fees, and legal or administrative costs associated with preparing influencer contracts (Alain, 2023). It varies with multiple factors such as marketers' locations, campaign goals, exclusivity, etc. We assume that c is uniformly distributed between 0 and $C > 0$ and, for ease of communication, we let $C = \frac{1}{2}$.[9] By collaborating with an influencer n with followers centered around x, a marketer with cost c obtains a revenue of $M(x, n)$ given below:

$$M(x,n) = \int_{x-\frac{n}{2}}^{x+\frac{n}{2}} \left(\max\left\{0, \alpha - |\tilde{x} - 0|_d\right\}\right) \rho d\tilde{x}, \tag{2}$$

where $\alpha > 0$ denotes the expected revenue from the target consumer and $|\tilde{x} - 0|_d$ denote the distance between influencer n's follower who is located at $\tilde{x}$ and marketer c's target consumer (who is located at 0). Specifically, $|\tilde{x}|_d \equiv \min\{1 - |\tilde{x}|, |\tilde{x}|\}$ according to the unit circle model. Hereafter, we refer to this revenue M as the "converted revenue" from influencer marketing campaigns. Without loss of

[9]Our key argument about the impact of matching accuracy on platform revenue is qualitatively invariant to the level of C, as proved in Appendix on page A11.

generality, we focus on the region where $\alpha \in (0, \frac{1}{4})$. This revenue function implies that the further away a follower of influencer n is located from the marketer's target consumer y, the lower converted revenue the marketer could extract from this follower, until it drops to zero.[10] Given the revenue function, we can characterize the profit of marketer c that purchases endorsement from influencer n with followers centered around x as follows:

$$U(x, n, c, p(n)) = M(x, n) - p(n) - c. \tag{3}$$

Let $\mathcal{C}$ denote the set of participating marketers who purchase endorsements from influencers. Given the proposed prices, a marketer would choose to participate, i.e., $c \in \mathcal{C}$, only if $U(\cdot) \geq 0$.

2.3 The Platform

The platform offers an AI-based matching service to increase the probability that marketers are matched with "suitable" influencers. Such a service is viable because the platform not only has access to influencers' metadata, but also the historical sales data generated by their marketing campaigns, and thus can build an AI/machine learning model to help match marketers with influencers. Mathematically, the matching AI recommends to each marketer a set of influencers $n \in [0, \frac{1}{2}]$ whose followers are centered around $\hat{x}$ given by

$$\hat{x} = 0 + \frac{\kappa}{2},$$

where $\kappa \in [-\frac{1}{2}, \frac{1}{2}]$ represents the bias of AI since the target consumer is located at 0.[11] That is, the AI-recommended influencers have a follower base located between $[\hat{x} - \frac{n}{2}, \hat{x} + \frac{n}{2}]$.[12] Figure 3 illustrates how the brand-influencer matching AI recommends influencers of different content broadness to marketers.

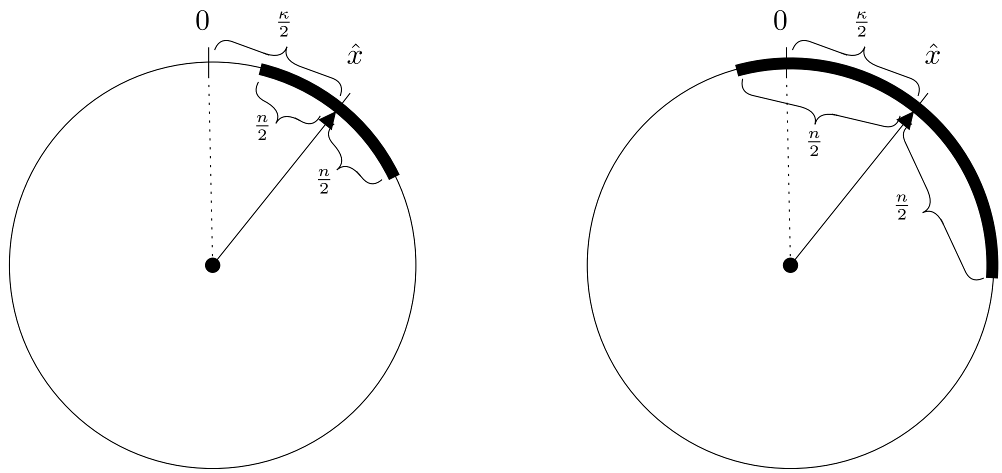
We define $\lambda = 1 - |\kappa| \in [\frac{1}{2}, 1]$ as the accuracy of the matching technology. Note that when $\lambda = 1$, the recommended influencers have followers whose interests are centered around marketer's target

[10]Without loss of generality, the "transportation cost" parameter (as in a classic Hotelling model) is normalized to 1 for ease of expression. This is an innocuous assumption that will not alter the qualitative results. In addition, this circular model allows us to study the alignment of followers and marketers' interests without getting into the complexities of asymmetry and corner solutions in a standard Hotelling model.

[11]Note that $\hat{x}$ and κ are one-to-one in our model. However, they are conceptually different variables. $\hat{x}$ refers to a location whereas κ refers to the inaccuracy of matching.

[12]This modeling choice for matching accuracy eliminates the heterogeneity in the core followers of competing influencers to keep the model simple and tractable. In Appendix A.2.3 on page A24, we show that our results remain qualitatively the same in an alternative model where this restriction is lifted.

Figure 3: Illustration of Brand-Influencer AI Recommendation



$n \leq |\kappa|$ $\qquad$ $n > |\kappa|$

Note: Marketers' target consumers are located at 0. The bold arc in each subfigure represents the followers of an AI-recommended influencer with a following size of n.

consumer, i.e. $\hat{x} = 0$. On the other extreme, when $\lambda = \frac{1}{2}$, the AI recommends influencers to a marketer such that the distance of the influencers' core follower and the marketer's target consumer is $\frac{1}{4}$, which is equivalent in expectation to the case where influencers are randomly matched with marketers because $\mathbf{E}[|\hat{x} - 0|_d] = \frac{1}{4}$.

Given the matching technology specified above, we can re-write the revenue function $M(\cdot)$ from a potential marketer-influencer match as follows:

$$M(\hat{x}, n) = \int_{\frac{\kappa}{2}-\frac{n}{2}}^{\frac{\kappa}{2}+\frac{n}{2}} \left(\max\{0, \alpha - |\tilde{x}|\}\right) \rho d\tilde{x}.$$

Note that for a given λ (or κ), the revenue function $M(\cdot)$ is independent of the exact location of influencers' core followers ($\hat{x}$) or that of the marketer's target consumers on the unit circle – it only varies with influencers' content broadness n. The following lemma gives the expression of the influencer marketing converted revenue $M(n)$.

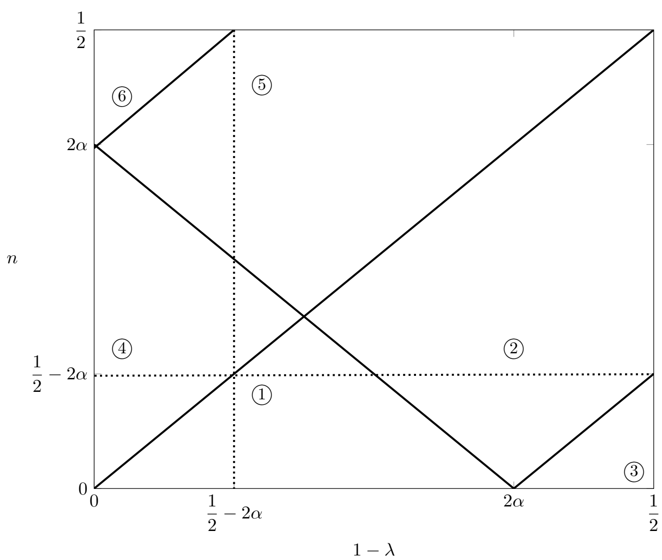
Lemma 1. *The converted revenue from influencer marketing, $M(\cdot)$, is given by*

$$M(n) = \rho m(n), \tag{4}$$

where

$$m(n) = \begin{cases} ① \; n\left(\alpha - \frac{1-\lambda}{2}\right) & \textit{if } n \leq 1-\lambda \textit{ and } n \leq 2\alpha-(1-\lambda), \\ ② \; \frac{(n-1+\lambda+2\alpha)^2}{8} & \textit{if } n \leq 1-\lambda \textit{ and } n > |2\alpha-(1-\lambda)|, \\ ③ \; 0 & \textit{if } n \leq 1-\lambda \textit{ and } n \leq (1-\lambda)-2\alpha, \\ ④ \; n\alpha - \frac{(1-\lambda)^2+n^2}{4} & \textit{if } n > 1-\lambda \textit{ and } n \leq 2\alpha-(1-\lambda), \\ ⑤ \; \frac{4\alpha^2+4\alpha(n-1+\lambda)-(n-1+\lambda)^2}{8} & \textit{if } n > 1-\lambda \textit{ and } 2\alpha-(1-\lambda) < n < 2\alpha+(1-\lambda), \\ ⑥ \; \alpha^2 & \textit{if } n > 1-\lambda \textit{ and } n \geq 2\alpha+(1-\lambda). \end{cases} \tag{5}$$

Figure 4: Regions of Converted Revenue $M(\cdot)$ vs. n and $1-\lambda$



Note: Circled numbers in the figure refer to the corresponding expressions in equation (5).

Figure 4 illustrates the six different regions of (λ, n) and the corresponding revenue function expressions. Given Lemma 1, we can suppress the input of x in the marketer's profit function as

$$U(n, c, p(n)) = M(n) - p(n) - c. \tag{6}$$

In other words, the marketer's expected profit only depends on the interest broadness (n) and density (ρ) of followers that will meet the marketer via the influencer, the price paid to the influencer $p(n)$, and the campaign fixed cost c.

The platform charges a percentage commission fee $\delta \in (0, 1)$ to all participating influencers and marketers for providing the marketplace of those marketing campaigns. This fee is considered to be the major source of the platform's revenue. Therefore, the platform's expected revenue function, π, given the influencers' proposed prices $p(n)$, is characterized by

$$\pi = \int_{n \in \mathcal{N}} \delta p(n) dn. \tag{7}$$

In the baseline model, we assume that this commission rate δ is exogenously given. In Section 4.2, we will discuss the case when δ is endogenously determined by the platform.

2.4 Timing and Equilibrium Concept

The timing of the game is as follows. First, given the level of matching accuracy λ, the platform's AI recommends a set of influencers to each marketer. Each influencer proposes a take-it-or-leave-it price for paid endorsement in a competitive market. Given the proposed prices, each marketer chooses whether to participate and buy paid endorsement from at most one influencer to maximize their profit. If a marketer buys endorsement from an influencer, the influencer creates and posts sponsored content, which generates converted revenue to the marketer based on the revenue function specified above. We refer to these influencers as "participating" influencers or influencers "in the market." If an influencer is not chosen by any marketer and thus not participating in influencer marketing campaigns, she obtains a reservation utility from her outside option. Finally, the platform and the participating influencers share the prices generated by those marketing campaigns. In our setup, an equilibrium is defined as follows.

Definition 1. *At any given λ, an equilibrium is a strategy profile such that:*

1. *$p^*(n)$ maximizes every influencer n's expected utility.*

2. *Every participating marketer $c \in \mathcal{C}$ that chooses influencer $n \in \mathcal{N}$ prefers buying an endorsement from her at $p^*(n)$ to buying from other influencers at their equilibrium prices; every non-participating marketer $c \notin \mathcal{C}$ obtains non-positive utility from buying an endorsement from any influencers at their respective prices.*

3. *The mass of participating influencers equals the mass of participating marketers, with each marketer matched with one influencer. That is, we have the following market clearing condi-*

tion:

$$\int_{n\in\mathcal{N}} dn = \int_{c\in\mathcal{C}} dc. \tag{8}$$

To refine our equilibria, we assume that non-participating influencers propose the lowest price when they are indifferent between proposing different prices to maximize their utility.[13] We start with studying the baseline model where the percentage commission fee (δ) is fixed.

3 Equilibrium Analysis

To simplify our analysis, we focus on the parameter space where the matching accuracy is high enough, i.e., $\lambda \in \left[\max\left\{\frac{1}{2}+2\alpha, 1-\alpha\right\}, 1\right]$, which corresponds to regions 1, 4, 5, and 6 in Figure 4. Given the matching accuracy λ, we first investigate how influencers' prices and marketers' participation decisions are determined in equilibrium, as described in the following lemma.

Lemma 2. ***(Equilibrium Characterization)*** *There exists a unique pair of thresholds* $(\overline{c}, \underline{n}) \in [0, \frac{1}{2}]^2$ *such that marketers and influencers participate if and only if* $c \in \mathcal{C} = [0, \overline{c}]$ *and* $n \in \mathcal{N} = [\underline{n}, \frac{1}{2}]$. *In this unique equilibrium, the following conditions are satisfied:*

- ***(Indifference Condition)*** *For any participating marketer* $c \le \overline{c}$, *they are indifferent between collaborating with any participating influencers of different sizes, i.e.,*

$$U(n_1, c, p^*(n_1)) = U(n_2, c, p^*(n_2)), \forall n_1, n_2 \in \mathcal{N}.$$

- ***(Price Condition)*** *For any participating influencer* $n > \underline{n}$, *we have* $p^*(n) = M(n) - \overline{c}$. *For the marginal participating influencer* $n = \underline{n}$ *and any non-participating influencer* $n < \underline{n}$, *we have* $p^*(n) = \frac{\gamma_0}{1-\delta}$.

Lemma 2 shows that the marginal participating marketer ($c = \overline{c}$) obtains zero profit from matching with any participating influencers at their proposed prices, and the marginal influencer's ($n = \underline{n}$) equilibrium price is equal to her "effective" opportunity cost ($\frac{\gamma_0}{1-\delta}$). According to Lemma 2 and the market clearing condition (8), it is easy to see that $(\overline{c}, \underline{n})$ must solve the following two

[13]This is an innocuous assumption to ensure the uniqueness of our equilibrium. It does not affect our results since the prices of non-participating influencers do not enter the platform's revenue function.

equations in equilibrium:

$$\frac{1}{2} - \underline{n} = \overline{c}, \tag{9}$$

$$M(\underline{n}) - \overline{c} = \frac{\gamma_0}{1-\delta}. \tag{10}$$

Hence the platform's equilibrium revenue, denoted by Π, can be simplified as follows:

$$\Pi = \int_{\underline{n}}^{\frac{1}{2}} \delta \underbrace{(M(n) - \overline{c})}_{p^*(n)} dn. \tag{11}$$

The following proposition characterizes how the platform's revenue changes with respect to the matching accuracy in equilibrium, which is illustrated by Figure 5 and Figure 6.

Proposition 1. ***(Impact of AI Improvement on Platform Revenue)*** *In equilibrium, the platform's revenue (Π) does not always increase in matching accuracy (λ). In particular, the point derivative of platform revenue with respect to matching accuracy is negative ($\frac{\partial \Pi}{\partial \lambda} < 0$), when the followers' density is intermediate within a certain range, i.e., $\rho \in (\underline{\rho}(\lambda), \overline{\rho}(\lambda))$. Otherwise, $\frac{\partial \Pi}{\partial \lambda} = 0$ when $\rho \leq \underline{\rho}(\lambda)$ or $\rho = \overline{\rho}(\lambda)$ and $\frac{\partial \Pi}{\partial \lambda} > 0$ when $\rho > \overline{\rho}(\lambda)$.*

Figure 5: Relationship between ρ and Point Derivative of Π w.r.t. λ

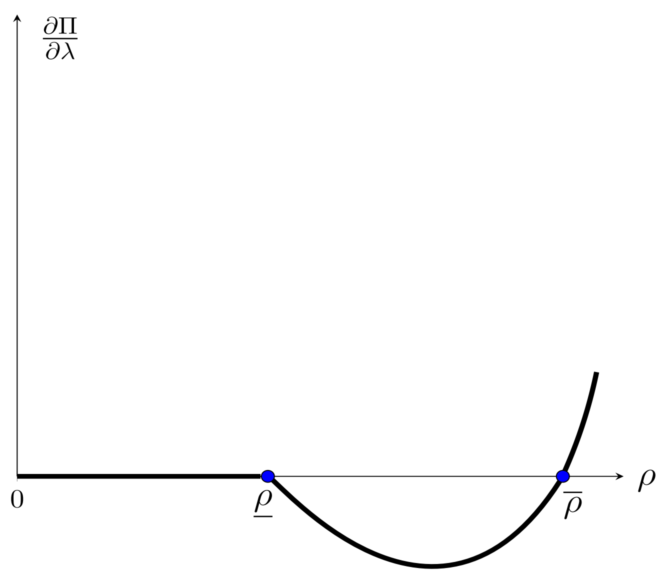
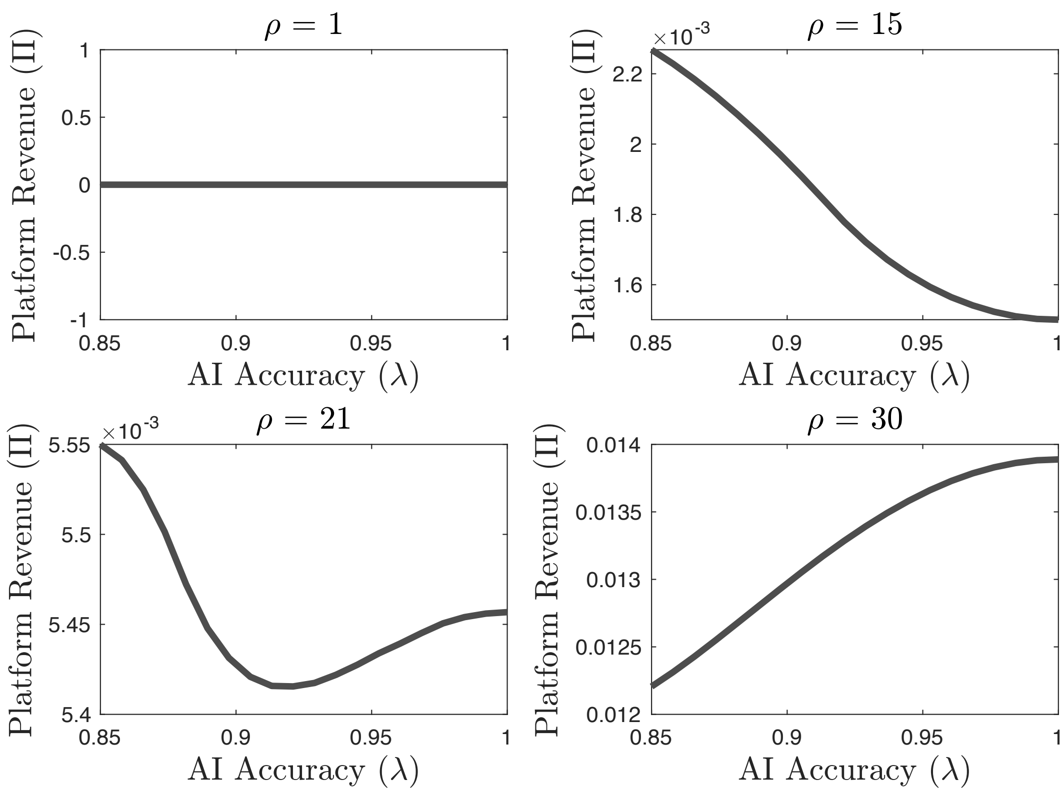


Figure 5 illustrates Proposition 1. We prove the existence of $\underline{\rho}$ and $\overline{\rho}$ in the Appendix. Note that Proposition 1 speaks to the point derivative $\frac{\partial \Pi}{\partial \lambda}$. For ease of understanding, we also provide

Figure 6 to show how the platform's revenue changes with matching accuracy at different values of ρ. As shown in the subfigures, when ρ is in some intermediate range, platform revenue Π can decrease in matching accuracy λ.

Figure 6: Impact of AI improvement on Platform Revenue ($\alpha = \delta = 0.15$)



Proposition 1 shows that the improvement of AI would hurt the platform's revenue under some circumstances. Interestingly, this implies that the platform may refrain from developing matching technology even if the cost of technology development is negligible. To provide some intuition, we can decompose the total impact of improving AI on the platform's revenue into the following three effects by taking the first-order derivative of Π with respect to λ, as described in expression (12):

$$\frac{\partial \Pi}{\partial \lambda} = \delta \left[\int_{\underline{n}}^{\frac{1}{2}} \underbrace{\left(\underbrace{\frac{\partial M}{\partial \lambda}}_{\text{matching enhancement } (+)} \underbrace{- \frac{\partial \overline{c}}{\partial \lambda}}_{\text{competition } (-)} \right)}_{\frac{\partial p^*(n)}{\partial \lambda}\text{: effect on equilibrium prices}} dn + \underbrace{\left(- \frac{\partial \underline{n}}{\partial \lambda} \right) \cdot p^*(\underline{n})}_{\text{market expansion } (+)} \right] \tag{12}$$

- *Matching enhancement effect.* The converted revenue from each marketer-influencer match is higher when influencers' followers are centered closer to the marketers' target audience, causing an upward effect on the influencers' equilibrium prices.

- *Competition effect.* When more influencers are matched with marketers, the extant influencers compete with new entrants[14] for marketers' attention, causing a downward pressure on the influencers' equilibrium prices.

- *Market expansion effect.* When more marketers buy endorsements from influencers, the platform obtains more revenue from more participation in equilibrium.

When the matching accuracy (λ) is higher, the followers of the AI-recommended influencers are centered closer to marketers' target consumers. This in turn enhances the converted revenue from every marketer-influencer match and reflects the marginal benefit of improving AI to the platform. Therefore, it provides an upward effect on influencers' equilibrium prices (the term $\frac{\partial M}{\partial \lambda}$ in expression (12)) and thus a positive effect on platform revenue. Meanwhile, more marketers have incentives to buy endorsements from influencers at their current prices as the converted revenue from doing so has increased. This leads to more niche influencers participating in marketing campaigns in equilibrium, which is captured by $\frac{\partial \underline{n}}{\partial \lambda}(< 0)$. This result echos the emerging trend of the "rise of the micro-influencers" where marketers seek after influencers with smaller audiences for endorsement (Fainmesser and Galeotti, 2021). Since more influencers participate, there is a competition effect that causes a downward pressure on influencer prices in equilibrium (the term $-\frac{\partial \bar{c}}{\partial \lambda}$ in expression (12)), which is a negative contribution to the platform's revenue. On the other hand, more influencers and marketers entering the market is associated with what we term the "market expansion effect." More influencers participating in the influencer marketing campaigns increases the total revenue that a platform could obtain from those campaigns (the term $\left(-\frac{\partial \underline{n}}{\partial \lambda}\right) \cdot p^*(\underline{n})$ in expression (12)). This positive market expansion effect is proportional to the price of the marginal influencer who is just indifferent between participating or not.[15]

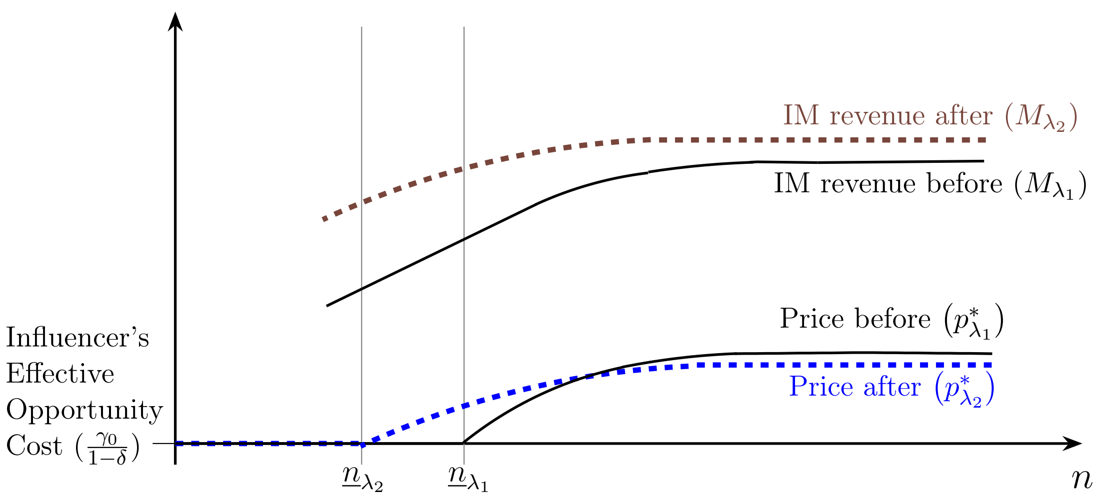
Since the total effect of AI improvement on equilibrium prices is composed of two countervailing effects — the positive matching enhancement effect ($\frac{\partial M}{\partial \lambda}$) and the negative competition effect ($-\frac{\partial \bar{c}}{\partial \lambda}$)

[14] We use the term "new entrants" or "newly-entering influencers" to refer to the influencers who switch from non-participating to participating in equilibrium.

[15] In the baseline model, the market expansion effect is absent because, according to Lemma 2, the equilibrium price of the marginal influencer equals her opportunity cost which is zero, i.e. $p^*(\underline{n}) = \frac{\gamma_0}{1-\delta} = 0$. In Sections 4.1 and 4.2, we will describe how the presence of such market expansion effect would attenuate the first two effects on the platform's revenue.

— the net effect of AI improvement on prices is ambiguous. Specifically, the matching enhancement effect is *asymmetric* for different n. Note that, however, our key results critically depend on the heterogeneity in influencers' content broadness, but not necessarily that in their follower size or density, as shown in Section 4.5. When n is large, the matching enhancement effect decreases in n. This is because general influencers already have a broad follower base, so the converted revenue from their influencer marketing campaigns is less sensitive to the matching accuracy. That means, the newly-entering niche influencers enjoy a higher matching enhancement effect from AI improvement than those very general influencers (i.e., influencers with large n) and thus provide a higher surplus to the marketers at their current proposed prices. Thus, in the new equilibrium, all the participating influencers have to adjust their prices to compete with those newly-entering niche influencers, which we refer to as the competition effect. As a result, the total effect of AI improvement on equilibrium prices can be negative for influencers with larger n, although it is positive for those with smaller n. This implies that improving AI could hurt the platform's revenue.

Figure 7: Mechanism Illustration



Note: $\lambda_1 < \lambda_2$. "IM" in this figure refers to influencer marketing.

In Figure 7, we illustrate the whole process of AI improvement by further decomposing it into two phases: in Phase 1, we hold influencers' proposed prices unchanged; in Phase 2, we allow for influencers to adjust their prices. Consider the case when matching accuracy is λ_1. Influencers with $n \geq \underline{n}_{\lambda_1}$ participate in influencer marketing campaigns and influencers' equilibrium prices are given by $p^*_{\lambda_1}(n)$ (and non-participating influencers charge their effective opportunity cost $\frac{\gamma_0}{1-\delta}$). The converted revenue from the marketing campaign with influencer n is $M_{\lambda_1}(n)$. Suppose that the matching accuracy now improves from λ_1 to λ_2, where $\lambda_2 > \lambda_1$. In Phase 1, suppose prices do not change (still being $p^*_{\lambda_1}$), then the converted revenue from influencer marketing increases due to AI improvement (from curve M_{λ_1} to M_{λ_2} in Figure 7). As a result, more marketers and influencers participate in the campaign since the marketer previously on the margin ($\bar{c}_{\lambda_1}$) now can obtain strictly positive profit. Therefore, the marginal influencer changes from $\underline{n}_{\lambda_1}$ to $\underline{n}_{\lambda_2}$, where $\underline{n}_{\lambda_2} < \underline{n}_{\lambda_1}$.

Next we enter Phase 2 where the price adjustment takes place. Due to the nature of the matching technology, AI improvement is more beneficial to influencers with smaller n. As is shown in Figure 7, the distance between M_{λ_1} and M_{λ_2} is larger for smaller n. Therefore, the new entrants who are niche influencers (with small n) become more attractive to marketers than those very general influencers (with large n) who were participating when matching accuracy was λ_1. As a result, those influencers with large n have to lower their prices in order to remain competitive in the market, whereas influencers with small n can afford to increase their prices. The equilibrium prices change from curve $p^*_{\lambda_1}$ to $p^*_{\lambda_2}$ in Figure 7 and the total revenue of the platform is proportional to the area below the price curve between $\underline{n}$ and $\frac{1}{2}$. Thus, the platform's revenue could decrease in the matching accuracy because of the decline in equilibrium prices for large n.

Our analysis above also reveals that, in a counterfactual case where the improvement of matching technology were to benefit incumbent influencers more than the newly-entering niche ones, which is not supported by our micro-founded model of matching technology, technology improvement could never hurt the platform's revenue. This is because, in this counterfactual case, newly-entering niche influencers would be less attractive to marketers compared to incumbent influencers, and thus incumbent influencers would raise rather than decrease their equilibrium prices in spite of competition from more participating influencers. Meanwhile, by Lemma 2, the new entrants' equilibrium prices were already their effective opportunity costs and thus cannot decrease further. That means the total effect of technology improvement on prices is always positive, and thus its effect on the platform revenue cannot be negative.

In sum, improving matching accuracy could reduce platform's revenue when the competition effect dominates the matching enhancement effect. Proposition 1 shows that it occurs only if the followers' density (ρ) is neither too large nor too small. This is because when ρ is sufficiently small, niche influencers have very few followers. Hence, marketers find it profitable to buy endorsements only from general influencers as their followers generate high enough converted revenue. Meanwhile, the return on a better influencer-marketer match diminishes for those general influencers who already have a diverse follower base and are insensitive to AI improvement. In that case, AI improvement does not affect the converted revenue of influencer campaigns or the marketers' participation decisions. As both the matching enhancement effect and the competition effect turn out to be zero, the platform's revenue is independent of AI improvement. In contrast, when ρ is sufficiently large, more marketers and influencers participate in equilibrium because the converted revenue from an influencer marketing campaign is proportional to ρ. That is, the participating influencers consist of more niche ones who will increase their equilibrium prices after AI improvement. Meanwhile, one can show that the competition effect is smaller ($\frac{\partial^2 \bar{c}}{\partial \lambda \partial \rho} < 0$) when ρ is sufficiently large and thus fewer general influencers will lower their prices. As a result, it is less likely that the total effect of price changes on the platform revenue is negative. Thus, only when followers' density is intermediate, the AI improvement may hurt the platform's revenue.

Our analysis so far also suggests the following managerial implication for platforms. Recall that the negative competition effect from improved matching is brought by the entry of niche influencers in the market. Thus, one potential way for the platform to neutralize this negative impact is to recommend only a subset of AI-selected influencers to marketers by filtering out niche influencers with n less than some threshold n_0. However, there are several caveats when considering this strategy. First, while neutralizing the competition effect, restricting the entry of niche influencers also deprives a platform of the potential revenue they could have obtained from them. Thus, a platform needs to optimally choose the threshold n_0 according to the matching accuracy λ (as well as the follower density ρ) to potentially benefit from this filtering action. Based on our analysis in Appendix A.2.1, the optimal n_0 is the solution to equation $1 - 2n_0 - M(n_0) = 0$. Thus, while the negative impact of competition could be mitigated from restricting the set of recommended influencers, completely reversing the negative impact requires the platform to have precise information about all the model parameters to optimally filter out some influencers. Second, this filtering action is non-trivial because it is not just about restricting the number, but more importantly the *type*, of recommended influencers, and thus the filtering could limit the growing potential of niche influencers despite its short-term benefit to the platform. Therefore, such practice could be subject to fairness concerns and limit the long-run development of the platform itself since it is not friendly to new, niche influencers. Third, if the marketers are fully aware of the platform's filtering strategy, they may bypass the platform's AI service to search for more cost effective influencers from, for example, third-party matching service providers. Since these transactions may take place off-the-platform, the platform loses the ability to benefit from the service commission and may subsequently suffer from a larger revenue loss.

4 Extensions and Discussions

4.1 Positive Opportunity Cost

To quickly deliver our key messages, so far we have been silent about two important aspects of the influencer marketing campaigns. One is the presence and heterogeneity of influencers' opportunity cost, and the other one is the platform's decision to adjust its commission fee. To explore the impact of each factor as well as their interaction, we add three extensions to our analysis progressively in Sections 4.1-4.3. First, we let the opportunity cost of influencers be a positive constant. Second, we allow the platform to adjust its commission fee in response to AI improvement. Third, we further allow the influencers' opportunity cost to be correlated with their number of followers.

In the baseline model, we have assumed away influencers' opportunity cost to focus on the main trade-off of AI improvement on the platform's revenue. In practice, influencers may need to devote a significant amount of time into running a marketing campaign via creating short videos or live-streaming. For example, a famous Chinese influencer Li Jiaqi, also known as "the No.1 seller of lipstick" with an annual income of more than \$1.4 million (CNY 10 million), is said to work long hours to prepare and manage influencer marketing campaigns and only have four hours of sleep a day during busy seasons.[16] In this section, we relax this assumption by allowing the opportunity cost to be positive, i.e., $\gamma_0 > 0$. The following proposition summarizes our analysis and shows that our results remain qualitatively the same as long as the influencers' opportunity cost is not too large.

Proposition 2. *When $\rho \geq \rho_0(\gamma_0) \equiv \frac{\gamma_0}{(1-\delta)\alpha^2}$, there exists $\overline{\gamma} > 0$ such that*

- *if $\gamma_0 < \overline{\gamma}$, we have*

$$\frac{\partial \Pi}{\partial \lambda} \begin{cases} = 0 & \text{if } \rho_0(\gamma_0) \leq \rho \leq \rho_1(\gamma_0) \text{ or } \rho = \rho_2(\gamma_0), \\ < 0 & \text{if } \rho_1(\gamma_0) < \rho < \rho_2(\gamma_0), \\ > 0 & \text{if } \rho > \rho_2(\gamma_0); \end{cases}$$

- *if $\gamma_0 \geq \overline{\gamma}$, the platform's revenue Π is non-decreasing in λ, i.e., $\frac{\partial \Pi}{\partial \lambda} \geq 0$.*

[16] https://pandayoo.com/2021/02/25/who-is-li-jiaqi-a-chinese-celebrity-with-6-hour-sales-of-600m-usd/.

When $\rho < \rho_0(\gamma_0)$, no influencers or marketers participate in influencer marketing campaigns in equilibrium and thus $\Pi(\lambda) \equiv 0$.

Proposition 2 shows that, when the influencers' opportunity cost is not too large, AI improvement would still hurt the platform's revenue if the influencers' follower density ρ is within a certain range. The intuition directly follows expression (12) but here the market expansion effect is positive since the price of the marginal participating influencer is no longer zero. According to the market clearing condition (8), the marginal change in the mass of marketers should equal to that in the mass of influencers, i.e., $\frac{\partial \bar{c}}{\partial \lambda} = -\frac{\partial \underline{n}}{\partial \lambda}$. Hence we can rewrite equation (12) as follows by collecting the last two terms:

$$\frac{\partial \Pi}{\partial \lambda} = \delta \int_{\underline{n}}^{\frac{1}{2}} \left(\underbrace{\frac{\partial M}{\partial \lambda}}_{\text{matching enhancement } (+)} \quad \underbrace{- \frac{\partial \bar{c}}{\partial \lambda}}_{\text{competition } (-)} \quad \underbrace{- \frac{\partial \underline{n}}{\partial \lambda} \left(\frac{p^*(\underline{n})}{\frac{1}{2} - \underline{n}} \right)}_{\text{market expansion } (+)} \right) dn \tag{13}$$

$$= \delta \left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M}{\partial \lambda} dn + \frac{\partial \bar{c}}{\partial \lambda} \left(-\bar{c} + \frac{\gamma_0}{1-\delta} \right) \right]. \tag{14}$$

Given that the matching enhancement effect remains positive for any $\gamma_0 > 0$, whether AI improvement hurts the platform's revenue critically depends on the net effect of the last two terms (the competition and market expansion effects). When the opportunity cost (γ_0) is sufficiently large, the market expansion effect can dominate the competition effect, i.e., $\frac{\gamma_0}{1-\delta} > \bar{c}$. This is because fewer marketers are willing to participate when the influencers charge higher prices that rise with their opportunity cost, so $\bar{c}$ is decreasing in γ_0. In that case, the platform's revenue always increases with a more accurate AI. Alternatively, when the opportunity cost is not too high, the negative competition effect could dominate the positive matching enhancement and market expansion effects if the followers' density (ρ) is neither too large nor too small, as discussed in the previous section.

4.2 Endogenous Commission Rate

So far, we have assumed that the revenue generated from an influencer's endorsement campaign is split between the influencer and the platform according to an exogenous commission rate δ. In practice, this is generally true in the short term. In the long run, however, the platform adjust the percentage fee it wants to charge from the campaign revenue. In other words, $\delta \in [0, 1]$ can be endogenously set by the platform. We discuss such possibility in this section and examine how it would affect the platform's revenue with the improvement of matching technology.

When setting its commission rate δ, the platform faces the following trade-off: on the one hand, higher δ implies higher revenue share for the platform; on the other hand, higher δ implies a lower revenue share for the influencers, forcing some to charge a higher price for the same campaign so that their surplus at least exceeds their opportunity cost.[17] This in turn deters some marketers from participating and consequently shrinks the total revenue the platform could obtain. Therefore, the platform needs to balance its revenue gain against marketers' participation constraints when setting its commission rate. Let $\delta^*(\lambda)$ denote the optimal commission rate at a given matching accuracy λ, and Π^* denote the platform's revenue when it sets the commission rate optimally. The following proposition summarizes how AI improvement affects the platform's revenue in this case.

Proposition 3. *(**AI Improvement with Endogenous Commission Rate**) If the platform can optimally choose its commission rate at any given matching accuracy, then the platform's revenue is non-decreasing in matching accuracy, i.e., $\frac{\partial \Pi^*}{\partial \lambda} \geq 0$.*

Proposition 3 suggests that adjusting the commission rate at different levels of matching accuracy allows the platform to always benefit from a better matching technology. This is because the platform now has the option to increase the commission rate so as to offset the negative competition effect whenever it dominates the positive market expansion effect associated with a better matching technology. That is, optimally adjusting the commission rate can mitigate the negative impact of AI improvement (although it may not fully eliminate the negative impact in a more general setting described in Section 4.3). Together with Proposition 1, these results imply that, in order to take advantage of a better AI, it is critical that the platform proactively sets its commission rate when the average followers' density falls within a certain range. In the following discussions, we investigate how the platform should adjust its service fee optimally under different circumstances.

Corollary 1. *At a given commission rate δ, if the platform's revenue declines with AI improvement λ, then it has the incentive to increase δ. Formally, we have*

$$\frac{\partial \Pi(\delta, \lambda)}{\partial \lambda} < 0 \Longrightarrow \frac{\partial \Pi(\delta, \lambda)}{\partial \delta} > 0.$$

Corollary 1 shows that, surprisingly, the platform's best response to a decline in revenue caused by AI improvement is to further increase its percentage commission fee. By taking the derivative of

[17] Note that when $\gamma_0 = 0$, the platform always prefers to charge as high commission fee as possible because it does not affect the influencers' equilibrium prices.

Π with respect to δ, we can summarize the effect of increasing the commission rate on the platform's revenue as follows:

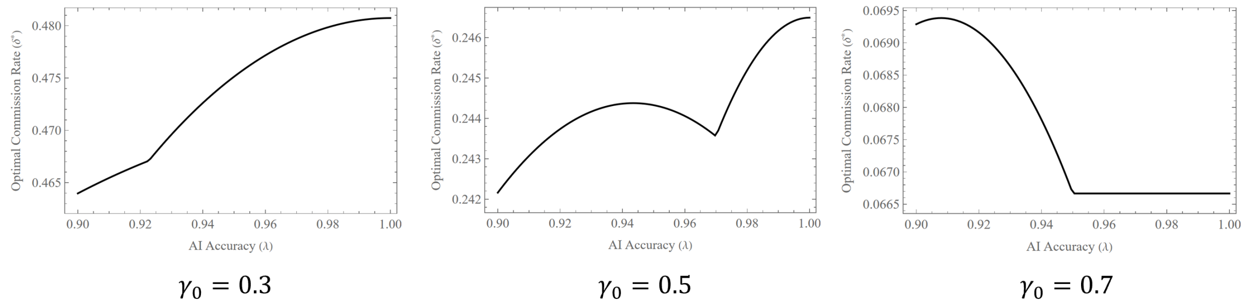
$$\frac{\partial \Pi}{\partial \delta} = \int_{\frac{1}{2}-\bar{c}}^{\frac{1}{2}} (M - \bar{c})\, dn + \delta \left[\frac{\partial \bar{c}}{\partial \delta} \left(-\bar{c} + \frac{\gamma_0}{1-\delta} \right) \right]. \tag{15}$$

The first term in expression (15) captures the marginal positive effect of increasing revenue share. In the second term of expression (15), the first component $\frac{\partial \bar{c}}{\partial \delta} (< 0)$ summarizes how the commission rate affects the market size in equilibrium – a higher δ implies a lower revenue share for the influencers, forcing some to charge a higher price for the same campaign and thus discourages some marketers from participating; the second component in the parenthesis $(-\bar{c} + \frac{\gamma_0}{1-\delta})$ corresponds to the net effect of competition and market expansion effects due to AI improvement (see equation (14)). Recall that, according to expression (14), improving matching accuracy may reduce platform's revenue only if the competition effect is sufficiently large that it dominates both the matching enhancement and the market expansion effect, implying that $(-\bar{c} + \frac{\gamma_0}{1-\delta})$ must be negative. In that case, increasing the commission rate δ always improves the platform's revenue as expression (15) becomes positive. In other words, if improving AI decreases the platform's revenue at a given commission rate, the platform has strict incentives to increase its commission rate. This strategy not only increases its revenue share but also effectively deters some marketers and influencers from participating and subsequently mitigates the negative competition effect.

At a first glance, one may presume that as matching accuracy improves, the platform should always increase its commission rate, given that the converted revenue is higher with a more accurate AI. However, this is not necessarily true—the platform may find it optimal to charge the influencers a lower commission rate with the improvement of AI. This is because a lower commission helps the platform facilitate the goal of market expansion and attract more influencers to participate in influencer marketing campaigns, albeit at the expense of a lower revenue share. This incentive is likely to dominate when the opportunity cost of influencers (γ_0) is high enough such that the participation rate is low in the first place and thus facilitating a larger market (as opposed to increasing revenue share) becomes the priority for a platform. Note that the relationship between the optimal commission rate and the accuracy of AI can be highly nonlinear[18], as shown in Figure 8. Hence, even though a platform could mitigate the adverse effects of AI advancements on its revenue by proactively adjusting its commission rate, this adjustment is not as straightforward as merely increasing the commission rate in tandem with technological improvements.

[18]Since solving the optimal commission rate is analytically challenging, we rely on numerical examples summarized in Figure 8 to illustrate the key point.

Figure 8: Optimal Commission Rate (δ^*) and Matching Accuracy (λ)



Note: $\alpha = 0.2$ and $\rho = 20$.

4.3 Opportunity Cost Related to Number of Followers

In the previous sections, we have assumed the opportunity cost to be constant for influencers of different sizes. However, since influencer marketing campaigns prevent influencers from engaging in and thus obtaining revenue from their followers in other businesses, influencers' opportunity cost may vary with the size of their followers. In this section, we explore how such opportunity cost affects the platform's revenue as well as its decision to adjust the commission rate.

Specifically, we assume that if an influencer does not participate in influencer marketing campaigns ($n \notin \mathcal{N}$), she obtains a reservation utility that is linear in her total number of followers (ρn). Put differently, influencer n's utility is now given by

$$V(p(n)) = \begin{cases} (1-\delta)\, p(n) & \text{if } n \in \mathcal{N}, \\ \gamma_0 + \gamma_1 \rho n & \text{if } n \notin \mathcal{N}, \end{cases} \tag{16}$$

where $\gamma_0 \geq 0$ and $\gamma_1 > 0$. We further assume $\gamma_1 < (1-\delta)(\alpha - \frac{1-\lambda}{2})$ to ensure the existence of an equilibrium.

Lemma 3. ***(Equilibrium Characterization)*** *There exists a triple of thresholds $(\overline{c}, \underline{n}, \overline{n}) \in [0, \frac{1}{2}]^3$ such that marketers and influencers participate if and only if $c \in \mathcal{C} = [0, \overline{c}]$ and $n \in \mathcal{N} = [\underline{n}, \overline{n}]$.*

Different from the baseline model, Lemma 3 implies that very general influencers with a diverse follower base ($n > \overline{n}$) would not participate in the influencer campaigns. This is because the marginal return falls below the marginal opportunity cost as n increases. Hence the platform's revenue can be written as

$$\Pi = \int_{\underline{n}}^{\overline{n}} \delta(M(n) - \overline{c})dn. \tag{17}$$

In the following proposition, we investigate whether the platform can fully eliminate the negative impact of AI improvement on its revenue by adjusting its commission rate.

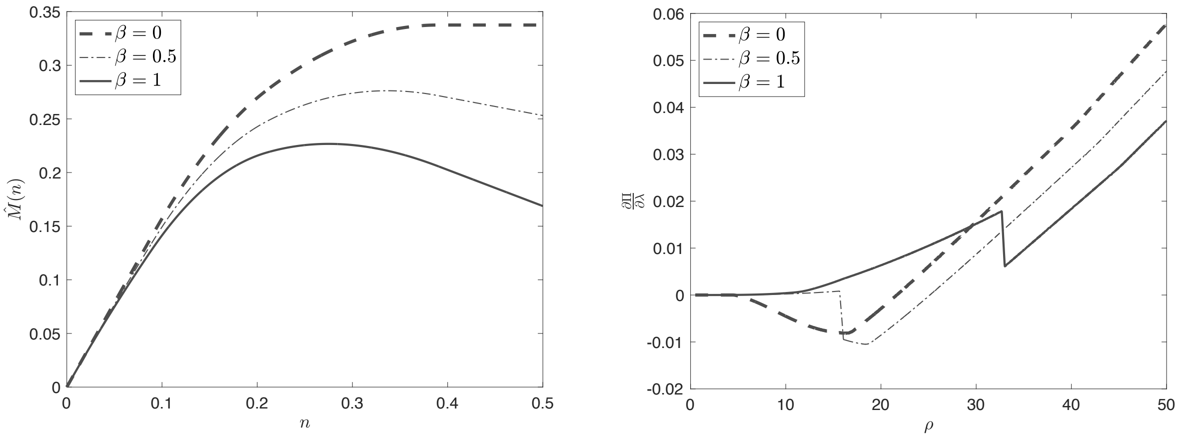
Proposition 4. *If influencers' opportunity cost is positively correlated with their number of followers, the platform's revenue may decrease in matching accuracy, whether or not the platform can optimally set its commission rate.*

Contrary to Proposition 3, Proposition 4 shows that allowing the platform to optimally adjust its percentage service fee could not eliminate the negative impact of AI improvement on the platform's revenue when influencers' opportunity cost increases in their size. To see this intuitively, we again decompose the effect of improving AI on the platform's revenue into the following three effects, as summarized in expression (18):

$$\frac{\partial \Pi}{\partial \lambda} = \delta \left[\int_{\underline{n}}^{\overline{n}} \left(\underbrace{\frac{\partial M}{\partial \lambda}}_{\text{matching enhancement } (+)} \underbrace{- \frac{\partial \overline{c}}{\partial \lambda}}_{\text{competition } (-)} \right) dn + \underbrace{\left(-\frac{\partial \underline{n}}{\partial \lambda} \right) \cdot p^*(\underline{n}) + \frac{\partial \overline{n}}{\partial \lambda} \cdot p^*(\overline{n})}_{\text{market expansion}} \right] \tag{18}$$

In this extended model, the market expansion effect now consists of two components: the first one is proportional to the equilibrium price of the most niche influencers who participate and the second one that of the most general influencers who participate. The second component is new compared to our analysis in Section 4.2, which reflects the effect of AI improvement on the maximum converted revenue obtained from an influencer-marketer match. Contrary to the case in Section 4.2, the presence of this effect makes increasing the percentage service fee δ particularly costly for the platform: a higher commission rate discourages not only the niche, but also general influencers from participating in equilibrium. The platform faces the following trade-off: on the one hand, it could mitigate the price competition effect by increasing the commission rate and deterring niche influencers from entering the market; on the other hand, it also risks losing revenue from crowding out the most general influencers when charging higher commission fees. In this case, commission adjustment ceases to be an effective remedy for the platform to fully offset the negative impact of AI improvement on revenue.

Our analysis in the extension sections above not only confirms the robustness of our main results, but also provides two managerial insights to platform owners and policymakers. First, the platform needs to proactively adjust its commission fee in order to take full advantage of or mitigate the damage from changes in matching technology. Second, in spite of its pricing autonomy



(a) $\hat{M}(n)$ vs. n (b) $\frac{\partial \Pi}{\partial \lambda}$ vs. ρ

Figure 9: Shape of $\frac{\partial \Pi}{\partial \lambda}$ at $\rho = 15$ and $\frac{\partial \Pi}{\partial \lambda}$ at $\alpha = 0.15$, $\lambda = 0.9$.

on commission, the platform may still suffer from a better brand-influencer AI especially when there is heterogeneity in the opportunity costs incurred by the influencer campaigns.

4.4 More Effective Niche Influencers

In our baseline model, we assume that influencers with different levels of content broadness have the same advertising effect on each follower. In practice, niche influencers could be more effective since their followers have more similar interests and they respond more actively to the influencers' endorsements. For example, a "ski influencer" may provide greater value to a "ski enthusiast" than does a "sports influencer."[19] In this section, we allow for this possibility by assuming that the expected revenue a brand obtains from collaborating with an influencer with content broadness n is given by

$$\hat{M}(n) = (1 - \beta n) M(n), \tag{19}$$

which means the original revenue $M(n)$ is discounted by $(1 - \beta n)$, where $0 \leq \beta \leq 2$. When $\beta = 0$, this model converges back to our baseline model. Figure 9a illustrates the shape of $\hat{M}(n)$ for $\beta = 0, 0.5$, and 1.

Similar to our analysis in the baseline model, the participation condition for an influencer would be $\hat{M}(n) \geq \bar{c}$, since marketers with lower cost c (i.e., $c \in [0, \bar{c}]$) would participate. Note that $\hat{M}(n)$ is now an "inverse-U" shaped function, as shown in Figure 9a. In other words, the converted revenue from collaborating with the most general influencers could be lower than some of the niche

[19] We thank an anonymous reviewer for pointing this out and suggesting this example.

ones. Thus, the set of participating influencers are not always those with large enough n as in the baseline model (i.e., $n \in [\underline{n}, \frac{1}{2}]$), but could possibly be those with intermediate n (i.e., $n \in [\underline{n}, \overline{n}]$) when the most general influencers become too ineffective in advertising.[20] Due to the mathematical complexity, we rely on a numerical exercise to illustrate the main findings. Figure 9b illustrates the relationship between the impact of matching accuracy on platform revenue ($\frac{\partial \Pi}{\partial \lambda}$) and the follower density (ρ) for $\beta = 0$, 0.5, and 1. It confirms the robustness of our main result that platform revenue could still decline in matching accuracy for intermediate follower density when β is not too large.

However, when the advertising effectiveness of an influencer is discounted by her content broadness, we see that the shape of $\frac{\partial \Pi}{\partial \lambda}$ with respect to ρ is different from that in the baseline model. Specifically, we have $\frac{\partial \Pi}{\partial \lambda} > 0$ and it increases in ρ when ρ is sufficiently small, then it discontinuously drops, and finally it increases in ρ again when ρ is sufficiently large. When ρ is sufficiently small, only the moderately general influencers (who provide the highest expected revenue to marketers) participate in equilibrium. Thus, the improvement of matching technology affects the participation decisions of both the most general ($\overline{n}$) and the most niche influencers ($\underline{n}$) on the margin, albeit in opposite directions. Given that the marginal general influencers are less sensitive to AI improvement than the marginal niche influencers, AI improvement further discourages those general influencers from participating whereas encouraging the niche ones to enter the market. This process could benefit the platform as it not only leads to a bigger market but also screens out the less profitable influencers in exchange for more profitable ones. When ρ exceeds a certain threshold such that the most general influencers (with $n = \frac{1}{2}$) also participate, however, there exists a "sudden drop" in the platform's equilibrium revenue. This is because, with sufficiently many followers generating sales, the most general influencers no longer exit the market as AI improves despite their advertising effectiveness being discounted by their generality. While still participating, they are pressured to make an even bigger cut in price, compared to our baseline model, because the competition effect is now exacerbated by their discounted advertising effectiveness, as the market encourages more niche influencers to enter.

On the one hand, when the gap is not too large between general and niche influencers' advertising effectiveness (i.e., β is not too large), the platform revenue could still decline in the matching accuracy as the competition effect, which could be even higher than that in the baseline model, dominates the market expansion and matching enhancement effects when the follower density ρ is intermediate. On the other hand, if β is sufficiently high, the platform revenue may never

[20] In some cases, it is still possible that $\overline{n} = \frac{1}{2}$ because $\hat{M}(n) \geq \overline{c}$ may hold for all $n \in [\underline{n}, \frac{1}{2}]$ if $\overline{c}$ is moderately small.

decline with improved matching accuracy, because the advertising effectiveness of the most general influencers is discounted too much by their generality that their contribution to the platform's revenue (and hence their price cut) is eventually outweighed by the contribution of other influencers who raise prices in equilibrium. This exercise suggests a boundary condition for our main results to hold: niche influencers are not far more effective than general ones.

4.5 Heterogeneity in Follower Size vs. Content Broadness

In the baseline model, we focus on modeling the heterogeneity in the content broadness (n) of influencers and assume away any heterogeneity in their follower density (ρ). This assumption implies a strictly positive correlation between the content broadness n and the influencers' size (ρn). To separately identify the role of heterogeneity in the influencers' size and the content broadness, as opposed to the baseline model where ρ is a constant, we consider two alternative models in this extension: (1) the influencers' size ρn is a constant and (2) the content broadness n is a constant.[21]

Model (1) First, we construct a model where all influencers have the same number of followers, that is, $\rho n \equiv N, \forall n \in (0, \frac{1}{2}]$, where $N > 0$. This setup implies a strictly negative correlation between influencers' content broadness and their follower density so that niche influencers have a higher follower density than general ones. In this case, we find that the platform revenue strictly increases in the matching accuracy λ if it is sufficiently high. Thus we extend our analysis to a wider range of λ, where $1 - 2\alpha \leq \lambda \leq 1$, so it could be lower than the possible values of λ in our baseline model. We show in the following proposition that, as in our baseline model, the platform's profit does not always increase with the improvement of matching accuracy between influencers and marketers.

Proposition 5. *If all influencers have the same number of followers N, there always exists $\underline{\lambda}(N)$ such that for $\lambda > \underline{\lambda}(N)$, the platform revenue ($\Pi$) first decreases and then increases in the matching accuracy (λ) in equilibrium. That is, $\frac{\partial \Pi}{\partial \lambda} < 0$ if $\lambda \in \left(\underline{\lambda}, \overline{\lambda}\right)$ and $\frac{\partial \Pi}{\partial \lambda} \geq 0$ if $\lambda \in \left(\overline{\lambda}, 1\right)$, where $\underline{\lambda} \equiv 1 - \frac{\left(N+4+2\sqrt{2}\right)\alpha}{\frac{N}{2}+4+2\sqrt{2}}$ and $\overline{\lambda} \equiv \min\left\{1 - \frac{\left(N+2+2\sqrt{2}\right)\alpha}{\frac{N}{2}+3+2\sqrt{2}}, 1-\alpha\right\}$.*

When all influencers have the same number of followers, the niche influencers provide sufficiently high revenue to marketers if the matching accuracy is sufficiently high ($\lambda > \overline{\lambda}$) that only influencers with a content broadness lower than a threshold ($n < \overline{n}$) participate in equilibrium, unlike in our

[21] In Appendix A.2.2 on page A24, we also provide a third model where both ρ and n are heterogeneous and ρn is not a constant to show the robustness of our results.

baseline model. In that case, market expansion from enhanced matching leads to the entrance of only general influencers. Since the niche influencers benefit more from enhanced matching compared to the general ones, all participating influencers now become more attractive to marketers than the marginal influencer with $n = \overline{n}$ at their original prices. Market clearing and indifference conditions imply that all participating influencers must have a strict incentive to increase their prices. Therefore, the platform's revenue must increase in the matching accuracy (λ). However, if the matching accuracy is not too high ($\underline{\lambda} < \lambda < \overline{\lambda}$)[22], not only some of the most general but also some of the most niche influencers do not participate in equilibrium because they could not provide more surplus to marketers than the lowest price they could afford to propose. In this case, market expansion from enhanced matching leads to more niche influencers entering the market because the positive impact of enhanced matching is higher on the niche influencers as opposed to general ones. As a result, the platform's revenue could decline as the entrance of niche influencers pressures the participating general influencers to lower their prices in equilibrium, just like in our baseline model.

This alternative model shows that our main result that the platform revenue could decline in matching accuracy is not driven by the fact that general influencers have more followers or that they all have to participate in the market.

Model (2) Second, we consider a model where all influencers have the same content broadness $n \equiv n_0 > 0$ whereas their follower density follows a uniform distribution, i.e., $\rho \sim U[0, \frac{1}{2}]$. In this case, it is straightforward to see that the matching enhancement effect from a better AI is the strongest for the largest influencers (with the highest follower density ρ). Given that the converted revenue $\rho m(n_0)$ increases in ρ, it is clear that only influencers with $\rho \geq \underline{\rho}$ would participate in equilibrium. The indifference condition implies that the participating influencers' prices are given by

$$p^*(\rho) = (\rho - \underline{\rho}) m(n_0), \tag{20}$$

where $m(n_0)$ is increasing in matching accuracy λ. The market clears when $\frac{1}{2} - \underline{\rho} = \overline{c}$ and the marginal marketer obtains zero profit, i.e., $\overline{c} = \underline{\rho} m(n_0)$. Therefore, the threshold $\underline{\rho}$ is given by $\underline{\rho} = \frac{1}{2(m(n_0)+1)}$. Since $m(n_0)$ increases in matching accuracy λ, we must have that $\underline{\rho}$ decreases in λ. Thus, enhanced matching leads to market expansion ($\underline{\rho}$ declines) and it empowers all participating influencers to raise their prices by equation (20). Since there is no negative competition effect on prices, the platform revenue always increases with the improvement of matching accuracy. This

[22]Note that $\overline{\lambda}$ converges to $\underline{\lambda}$ when N approaches 0 or infinity.

alternative model shows that our main result would not occur in the absence of heterogeneity in influencers' content broadness. This is because the impact of matching technology on influencers is now uniformly scaled by their follower size (or density).

From our baseline model as well as models (1) and (2) in this extension, we can see that our main mechanism critically depends on the existence of heterogeneity in influencers' content broadness, but not necessarily heterogeneity in their follower size or density. This is because the heterogeneity in influencer content broadness induces an asymmetric impact of enhanced matching more favorably on the newly-coming influencers from market expansion, which could hurt the platform's revenue through the negative competition effect on equilibrium prices.

4.6 Discussions on Competition

Our baseline model focuses on the competition among influencers for collaboration with marketers with the same target consumers. For tractability, however, we have abstracted away from modeling competition among marketers for consumer demand. To incorporate this marketer-level competition, we could modify our baseline model by adding a discount factor to the marketer's expected revenue from influencer marketing, such that the expected revenue decreases in the *equilibrium* intensity of influencer marketing. We analyze this modified model in Appendix A.2.4.1 and show that the impact of matching technology on platform revenue with respect to the follower density ρ is similar to that in our baseline model, where the platform revenue could decline in matching accuracy for intermediate ρ. However, the competition among brands for consumer demand dilutes the positive matching enhancement effect. This dilution effect mainly comes from three sources. First, for every participating marketer, the benefit of a higher matching accuracy on its revenue is discounted by its competitors targeting the same consumer segment. Second, this discount factor (a smaller discount factor reflecting a higher level of competition) itself declines in the number of competitors as improved matching attracts more marketers to participate in equilibrium. Third, due to the discounted revenue from influencer marketing, fewer marketers in total have incentives to collaborate with influencers which lead to fewer participating influencers. On the other hand, the total competition effect may be higher or lower than that in the baseline model, depending on the level of follower density ρ. Thus, the total effect of improved matching on platform revenue could be strengthened or weakened compared to our baseline model.

Besides, the competition among influencers is also limited in our baseline model because each influencer is restricted to collaborate with up to one marketer. This assumption is reasonable when marketers compete with each other. Nevertheless, if marketers do not compete, then it is possible that influencers offer their services to multiple marketers. We investigate this possibility progressively in Appendix A.2.4.2. First, we relax the assumption that each influencer can only collaborate with one marketer, while maintaining the assumption that all marketers have the same target consumers. We show that the equilibrium prices of all influencers decline with the number of marketers each influencer can collaborate with. Regardless, our analysis remains qualitatively the same because this price drop effect is orthogonal to AI improvement, as long as all types of influencers are endowed with equal opportunities to collaborate with multiple marketers. Second, it is possible that general influencers can collaborate with more marketers than niche ones, because general influencers can promote multiple brands with different target consumers due to their diverse follower base. In this case, the asymmetric impact of AI improvement on different types of influencers does interact with the different number of marketers each influencer can collaborate with. It is analytically challenging to extend our framework to model influencers promoting multiple marketers with different target consumers. Hence, we present a simple model with one general influencer and two niche influencers in Appendix A.2.4.2 to discuss the main insights. When matching accuracy increases, the general influencer has to lower her equilibrium price due to the competitive pressure from more suitable niche influencers in one product category. Furthermore, as the price in this product category declines, the general influencer has an incentive to serve more product categories by further decreasing her prices. In other words, enhanced matching leads to higher within-product-category competition among influencers and it spills over to competition in other product categories as well, exacerbating the potential decline in platform revenue.

5 Conclusion

Influencer marketing has become one of the hottest trending marketing strategies in recent years (Neate, 2020). A pain point for marketers interested in implementing influencer marketing is to find the "right" influencers to work with. One solution to this problem is for platforms to offer artificial intelligence (AI) to connect marketers with influencers. We build a theoretical model where marketers are matched with influencers on a social media platform through AI-enabled matching technology. Improved matching technology leads to better matches, generating higher converted revenue and prompting influencers to charge higher prices. However, due to the asymmetric impact of AI improvement on different types of influencers, enhanced matching also increases competition, causing some influencers to lower prices. The balance between these effects determines the platform's revenue, revealing a non-linear relationship between platform revenue and matching accuracy. In addition, follower density moderates this trade-off, influencing the platform's technology strategy.

Our study offers three key insights for social media platforms engaging in influencer marketing. First, while AI can improve influencer-marketer matching, it may sometimes backfire. Second, platforms with a specific influencer-follower composition may be less inclined to adopt AI. We find that when the platform is of moderate popularity, i.e., when follower density is intermediate, the platform may prefer not to adopt a better AI, even if the implementation cost is negligible. Third, to take full advantage of, or to mitigate the sometimes negative impact of AI improvement, platforms should always integrate its technology strategy with its pricing strategy by adjusting commission fees accordingly.

There are several ways to extend this line of research. First, one can explore a dynamic setting where influencers participate in recurring influencer marketing campaigns and are allowed to choose their positioning as niche or general influencers in response to a platform's AI improvement. Second, in our model, influencers propose a take-it-or-leave-it offer to marketers. Although this modeling choice is consistent with the context of paid endorsement with non-celebrity influencers, celebrity influencers may have the power to negotiate prices with marketers. Future research can investigate how this negotiation power changes the dynamics of competition among influencers on social media. Third, although our framework with a continuum of influencers and marketers integrates the key elements of this influencer market—influencers, brands, consumers, as well as the matching AI—in an elegant way, it is less suitable for studying the potential competition among brands with different target segments. Fourth, our model captures an important reason for platform users to purchase endorsed items: their interests are aligned with interests of the products' target customers. Future research can explore how other motives, such as influencers' low-price-guarantee, may also drive users' purchase. Finally, our results point out the dangers of using matching technologies without adjusting some platform features, such as the number and type of recommended influencers. We leave the formal analysis of this adjustment and its implications for future research. Future research could empirically test whether platforms using more accurate matching technologies also restrict the number and type of recommended influencers.

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All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The authors have no funding to report.

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APPENDIX

A.1 Proofs

Proof of Lemma 1. Without loss of generality, we can assume the bias, κ, is always positive. Based on Figure 3, we separately discuss the cases when $n \leq \kappa$ and when $n > \kappa$.

Case 1. $n \leq \kappa$**.** Within this case, there are three subcases based on whether the distance between the target consumer and the left (right) end of the recommended influencer's followers is less than α:

- if $\frac{\kappa}{2} + \frac{n}{2} \leq \alpha$, i.e., $n \leq 2\alpha - \kappa$, then $m(n) = \int_{\frac{\kappa}{2}-\frac{n}{2}}^{\frac{\kappa}{2}+\frac{n}{2}} (\alpha - z)dz = n(\alpha - \frac{\kappa}{2})$;
- if $\frac{\kappa}{2} + \frac{n}{2} > \alpha > \frac{\kappa}{2} - \frac{n}{2}$, i.e., $n > |2\alpha - \kappa|$, then $m(n) = \int_{\frac{\kappa}{2}-\frac{n}{2}}^{\alpha} (\alpha - z)dz = \frac{(n-\kappa+2\alpha)^2}{8}$;
- if $\frac{\kappa}{2} - \frac{n}{2} \geq \alpha$, i.e., $n \leq \kappa - 2\alpha$, then $m(n) = 0$.

Case 2. $n > \kappa$**.** Within this case, there are also three subcases:

- if $\frac{\kappa}{2} + \frac{n}{2} \leq \alpha$, i.e., $n \leq 2\alpha - \kappa$, then $m(n) = \int_0^{\frac{n}{2}-\frac{\kappa}{2}} (\alpha - z)dz + \int_0^{\frac{\kappa}{2}+\frac{n}{2}} (\alpha - z)dz = n\alpha - \frac{\kappa^2+n^2}{4}$;
- if $\frac{\kappa}{2} + \frac{n}{2} > \alpha > \frac{n}{2} - \frac{\kappa}{2}$, i.e., $2\alpha + \kappa > n > 2\alpha - \kappa$, then $m(n) = \int_0^{\frac{n}{2}-\frac{\kappa}{2}} (\alpha - z)dz + \int_0^{\alpha} (\alpha - z)dz = \frac{4\alpha^2+4\alpha(n-\kappa)-(n-\kappa)^2}{8}$;
- if $\frac{n}{2} - \frac{\kappa}{2} \geq \alpha$, i.e., $n \geq 2\alpha + \kappa$, then $m(n) = 2\int_0^{\alpha} (\alpha - z)dz = \alpha^2$.

Substituting κ with $1 - \lambda$ gives the desired expression of $m(n)$ in Lemma 1.

$\square$

Proof of Lemma 2. We first show that any strategy profile that constitutes an equilibrium must satisfy the indifference condition. That is, we want to show that there exists a profitable deviation if this condition does not hold: $U(n_1, c, p^*(n_1)) = U(n_2, c, p^*(n_2))$, $\forall n_1, n_2 \in \mathcal{N}$. Suppose there exist two influencers n_1 and n_2 of different sizes where n_1 is matched to marketer c_1 and n_2 to marketer c_2 in equilibrium. Meanwhile, suppose that marketer c_2 strictly prefers to collaborate with n_2 than with n_1, that is,

$$U(n_2, c_2, p^*(n_2)) > U(n_1, c_2, p^*(n_1)).$$

Then by equation (6), we have

$$M(n_2) - p^*(n_2) - c_2 > M(n_1) - p^*(n_1) - c_2,$$

which implies

$$M(n_2) - p^*(n_2) > M(n_1) - p^*(n_1),$$

and thus

$$U(n_2, c_1, p^*(n_2)) = M(n_2) - p^*(n_2) - c_1 > M(n_1) - p^*(n_1) - c_1 = U(n_1, c_1, p^*(n_1)).$$

By proposing a slightly higher-than-equilibrium price $p'(n_2) > p^*(n_2)$ such that $U(n_2, c_1, p'(n_2)) > U(n_1, c_1, p^*(n_1))$ still holds, influencer n_2 would still have someone to collaborate with (marketer c_1) and become strictly better off since the price is higher. This contradicts the notion of an equilibrium.

Next, given the necessity of the indifference condition, we prove the equilibrium structure and the price condition. Note that individual rationality implies that all participating influencers and marketers must be obtaining at least their outside utility. That is,

$$U(n, c, p(n)) = M(n) - p(n) - c \geq 0$$
$$V(p(n)) = (1 - \delta)p(n) \geq \gamma_0$$

On the one hand, a marketer's profit function $U(n, c, p(n)) = M(n) - p(n) - c = R - c$, where R is a constant by the indifference condition, strictly declines in c. Note that if $U(n, c, p(n)) = R - c < 0$ for all c, then any influencer can decrease her price to get matched to a marketer with $c = 0$ and get strictly better off. If $U(n, c, p(n)) = R - c > 0$ for all c, then all marketers participate, and thus all influencers participate due to the third condition in Definition 1. Consider $c = \frac{1}{2}$. We have $U(n, \frac{1}{2}, p(n)) = M(n) - p(n) - \frac{1}{2} > 0$ for any n, and thus $M(n) > p(n) + \frac{1}{2} \geq \frac{1}{2}$ for any n, which is clearly not true since $M(0) = 0$ and $M(\cdot)$ is continuous. Hence, by the monotonicity of $U(n, c, p(n))$ in c, there exists a marginal marketer $\bar{c} \in [0, \frac{1}{2}]$ such that $U(n, \bar{c}, p(n)) = 0$, and all marketers with $c < \bar{c}$ would strictly prefer collaborating with some influencer n to not participating as $U(n, c, p(n)) > U(n, \bar{c}, p(n)) = 0$. Furthermore, for the marginal marketer $\bar{c}$, by the indifference condition, we have in equilibrium

$$U(n, \bar{c}, p^*(n)) = M(n) - p^*(n) - \bar{c} = 0 \tag{A1}$$

for any n, and thus

$$p^*(n) = M(n) - \bar{c}.$$

On the other hand, given the indifference condition, for any $n > \underline{n}$ we have

$$M(n) - p^*(n) - c = M(\underline{n}) - p^*(\underline{n}) - c,$$

and thus

$$p^*(n) - p^*(\underline{n}) = M(n) - M(\underline{n}).$$

Because $\frac{\partial M}{\partial n} \geq 0$, we know that $M(n) \geq M(\underline{n})$ and thus $p^*(n) \geq p^*(\underline{n})$, or $p^*(n)$ is an increasing function of n. Therefore, similar to the existence of a marginal marketer, there exists a marginal influencer $\underline{n}$ such that $(1-\delta)p^*(\underline{n}) = \gamma_0$, and any influencer with $n > \underline{n}$ would also participate because they obtain a utility higher than their outside utility ($p^*(n) \geq p(\underline{n}) = \frac{\gamma_0}{1-\delta}$). For non-participating influencers, they are indifferent between setting any prices higher than $\frac{\gamma_0}{1-\delta}$. By the assumption that influencers propose the lowest price when they are indifferent between proposing different prices, we have $p^*(n) = \frac{\gamma_0}{1-\delta}$ for $n < \underline{n}$.

Finally, we confirm that a strategy profile is indeed an equilibrium when the two conditions hold. Marketers and non-participating influencers obviously have no incentives to deviate. For any participating influencer $n_1 \in \mathcal{N}$, we must have $V(p^*(n_1)) \geq \gamma_0$. She would not have an incentive to propose a lower price because it only yields a lower utility. Neither does she have an incentive to propose a higher price such that $p'(n_1) > p^*(n_1)$. This is because, given the indifference condition, we must have $U(n_1, c, p'(n_1)) < U(n_1, c, p^*(n_1)) = U(n_2, c, p^*(n_2))$ for any $n_2 \in \mathcal{N}\backslash\{n_1\}$ and c. As a result, no marketer would choose to collaborate with influencer n_1 at price $p'(n_1)$, and thus influencer n_1 would obtain her outside utility γ_0 which is weakly lower than $V(p^*(n_1))$. Note that a single-point deviation of n_1 will not result in any marketer left unmatched due to the uncountable nature of $\mathcal{M}$ and $\mathcal{N}$. Hence, there is no profitable deviation for any participating influencer. For any non-participating influencer n, we must have $p^*(n) < \frac{\gamma_0}{1-\delta}$. Her utility would remain unchanged unless she lowers her proposed price to $p' < p^*(n)$ which implies that she will be strictly preferred by all marketers. However, participating in a marketing campaign at price p' renders her lower utility than her outside options as $p' < p^*(n) < \frac{\gamma_0}{1-\delta}$. Therefore, there is no profitable deviation either for any non-participating influencer.

$\square$

Proof of Statement in Footnote 8. If we allow for influencers to take a commission proportional to their sales, then the total price that an influencer with content broadness n can collect from a marketer, $p(n)$, is composed of a fixed fee, denoted by $p_0(n)$, and the commission which is proportional to the sales generated by the influencer marketing campaign, $M(n)$. In other words, the total price $p(n)$ is given by

$$p(n) = p_0(n) + \eta M(n),$$

where $\eta \in [0,1]$. $p_0(n)$ is the decision variable chosen by the influencer. Denote $p_0(n)$ at the equilibrium level as $p_0^*(n)$. Note that the indifference condition described in Lemma 2 clearly still holds, since the same logic still applies. We only need to show that the price condition also holds, then it is sufficient to conclude that our results would remain the same in the subsequent propositions.

Clearly, for any non-participating influencers ($n < \underline{n}$), they will charge $p^*(n) = \frac{\gamma_0}{1-\delta}$ by the assumption that influencers propose the lowest price when they are indifferent between different prices. For the participating influencers, the indifference condition implies equation (A1), which in this case becomes

$$U(n, \overline{c}, p^*(n)) = M(n) - (p_0^*(n) + \eta M(n)) - \overline{c} = 0,$$

This implies $p_0^*(n) = (1-\eta)M(n) - \overline{c}$, and thus

$$p^*(n) = p_0^*(n) + \eta M(n) = M(n) - \overline{c},$$

which is exactly the same as the price condition in Lemma 2.

□

Proof of Proposition 1. We solve for $(\overline{c}, \underline{n})$ in equilibrium by solving the system of equations (9) and (10), which is equivalent to solving equation (A2) for $\overline{c}$ below.

$$\rho \cdot m\left(\frac{1}{2} - \overline{c}\right) - \overline{c} = t, \tag{A2}$$

where $t \equiv \frac{\gamma_0}{1-\delta}$. In the main model, we have assumed that $\gamma_0 = 0$ so $t = 0$. Under the assumption that $\lambda \in \left[\max\left\{\frac{1}{2} + 2\alpha, 1-\alpha\right\}, 1\right]$, we can reduce the expression of $m(\cdot)$ given by equation (5) to

$$m(n) = \begin{cases} n\left(\alpha - \frac{1-\lambda}{2}\right) & \text{if } n \le 1-\lambda, \\ n\alpha - \frac{(1-\lambda)^2 + n^2}{4} & \text{if } 1-\lambda < n \le 2\alpha - (1-\lambda), \\ \frac{4\alpha^2 + 4\alpha(n-1+\lambda) - (n-1+\lambda)^2}{8} & \text{if } 2\alpha - (1-\lambda) < n < 2\alpha + (1-\lambda), \\ \alpha^2 & \text{if } n \ge 2\alpha + (1-\lambda). \end{cases} \tag{A3}$$

Denote LHS of equation (A2) as $G(\overline{c})$. $G(\overline{c})$ is clearly decreasing in $\overline{c}$, since $\rho m(\cdot) = M(\cdot)$ is an increasing function by Lemma 1. In addition, $G(\frac{1}{2}) = -\frac{1}{2} < t$ and $G(0) = \rho\alpha^2 \ge t = 0$. Thus, there is a unique solution for $\overline{c}$ to equation (A2), and thus a unique solution for $(\overline{c}, \underline{n})$ in equilibrium.

Before solving for $\underline{n}$ and $\overline{c}$, we first derive the expression for $\frac{\partial \Pi}{\partial \lambda}$. By equation (11), we have

$$\begin{aligned} \Pi &= \int_{\underline{n}}^{\frac{1}{2}} \delta\left(M(n) - \overline{c}\right) dn \\ &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} M(n) dn - (\frac{1}{2} - \underline{n})\overline{c}\right] \\ &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} M(n) dn - \overline{c}^2\right]. \end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{\partial \Pi}{\partial \lambda} &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda} dn - \frac{\partial \underline{n}}{\partial \lambda} M(\underline{n}) - 2\overline{c}\frac{\partial \overline{c}}{\partial \lambda}\right] \\
&= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda} dn + \frac{\partial \overline{c}}{\partial \lambda}(M(\underline{n}) - 2\overline{c})\right] \\
&= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda} dn + \frac{\partial \overline{c}}{\partial \lambda}(t - \overline{c})\right] \\
&= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \rho\frac{\partial m(n)}{\partial \lambda} dn + \frac{\partial \overline{c}}{\partial \lambda}(t - \overline{c})\right], \qquad \text{(A4)}
\end{aligned}$$

where the second equality is due to $\overline{c} = \frac{1}{2} - \underline{n}$, and the third equality is because $M(\underline{n}) - \overline{c} = p^*(\underline{n}) = \frac{\gamma_0}{1-\delta} = t$ by Lemma 2. Thus,

$$\text{sign}\left(\frac{\partial \Pi}{\partial \lambda}\right) = \text{sign}\left[\int_{\underline{n}}^{\frac{1}{2}} \rho\frac{\partial m(n)}{\partial \lambda} dn + \frac{\partial \overline{c}}{\partial \lambda}(t - \overline{c})\right]. \qquad \text{(A5)}$$

For ease of expression and analysis, we let $v \equiv \frac{1}{\rho}$, and thus $M(\cdot) = \frac{m(\cdot)}{v}$. Since $M(\cdot)$ and $m(\cdot)$ are piece-wise functions, we need to have the following discussion about the range of $\frac{1}{2} - \overline{c}$:

Case 1. Condition: $\frac{1}{2} - \overline{c} \geq 2\alpha + (1 - \lambda)$.

In this case, $m(\frac{1}{2} - \overline{c}) = \alpha^2$, and equation (A2) is reduced to $\frac{\alpha^2}{v} - \overline{c} = t$. Therefore,

$$\overline{c} = \frac{\alpha^2}{v} - t,$$

and

$$\underline{n} = \frac{1}{2} - \overline{c} = \frac{1}{2} - \frac{\alpha^2}{v} + t.$$

The condition for this case $\frac{1}{2} - \overline{c} \geq 2\alpha + (1 - \lambda)$ is thus rewritten as

$$v \geq \frac{\alpha^2}{\lambda - \frac{1}{2} - 2\alpha + t}.$$

Note that the denominator $\lambda - \frac{1}{2} - 2\alpha + t$ is always non-negative since we have assumed that $\lambda \geq \frac{1}{2} + 2\alpha$.

The platform's revenue is thus given by

$$\Pi = \int_{\underline{n}}^{\frac{1}{2}} \delta\left(M(n) - \overline{c}\right) dn = \int_{\frac{1}{2} - \frac{\alpha^2}{v} + t}^{\frac{1}{2}} \delta\left(\frac{\alpha^2}{v} - \frac{\alpha^2}{v} + t\right) dn = \delta t(\frac{\alpha^2}{v} - t).$$

Clearly, $\frac{\partial \Pi}{\partial \lambda} = 0$, i.e., Π is independent of λ.

Case 2. Condition: $2\alpha - (1-\lambda) < \frac{1}{2} - \bar{c} < 2\alpha + (1-\lambda)$.

In this case, $m(\frac{1}{2}-\bar{c}) = \frac{4\alpha^2+4\alpha(\frac{1}{2}-\bar{c}-1+\lambda)-(\frac{1}{2}-\bar{c}-1+\lambda)^2}{8}$, and equation (A2) now becomes

$$\frac{4\alpha^2+4\alpha(\frac{1}{2}-\bar{c}-1+\lambda)-(\frac{1}{2}-\bar{c}-1+\lambda)^2}{8v} - \bar{c} = t,$$

which is equivalent to the following quadratic equation:

$$F(\bar{c}) \equiv \bar{c}^2 + (4\alpha+1-2\lambda+8v)\bar{c} - 4\alpha^2 + \alpha(2-4\lambda) - (1-\lambda)\lambda + 8tv + \frac{1}{4} = 0. \tag{A6}$$

Since we have the condition that $\frac{1}{2}-\bar{c} < 2\alpha + (1-\lambda)$, or $\bar{c} > -2\alpha - \frac{1}{2} + \lambda$, only the larger root of equation (A6) is a solution, and thus

$$\bar{c} = -2\alpha - \frac{1}{2} + \lambda - 4v + 2\sqrt{2\alpha^2 - 2tv + 4v^2 + 4\alpha v - 2\lambda v + v}. \tag{A7}$$

Therefore, the condition for Case 2, $2\alpha - (1-\lambda) < \frac{1}{2} - \bar{c} < 2\alpha + (1-\lambda)$, can be rewritten as

$$\frac{2\alpha^2 - (1-\lambda)^2}{3 - 4\alpha - 2\lambda + 2t} < v < \frac{\alpha^2}{\lambda - \frac{1}{2} - 2\alpha + t}.$$

By the implicit function theorem (IFT),

$$\frac{\partial \bar{c}}{\partial \lambda} = -\frac{F_\lambda}{F_{\bar{c}}} = \frac{\bar{c} + 2\alpha + \frac{1}{2} - \lambda}{4v + \bar{c} + 2\alpha + \frac{1}{2} - \lambda} > 0,$$

since $\bar{c} > -2\alpha - \frac{1}{2} + \lambda$ (which also implies $F_{\bar{c}} > 0$). By equation (A4), we have

$$\begin{aligned}
\frac{\partial \Pi}{\partial \lambda} &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda} dn + \frac{\partial \bar{c}}{\partial \lambda}(t-\bar{c})\right] \\
&= \delta\left[\frac{1}{v}\left(\int_{\underline{n}}^{2\alpha+(1-\lambda)} \frac{\partial \frac{4\alpha^2+4\alpha(n-1+\lambda)-(n-1+\lambda)^2}{8}}{\partial \lambda} dn + \int_{2\alpha+(1-\lambda)}^{\frac{1}{2}} \frac{\partial \alpha^2}{\partial \lambda} dn\right) + \frac{\partial \bar{c}}{\partial \lambda}(t-\bar{c})\right] \\
&= \delta\left[\frac{1}{v}\int_{\frac{1}{2}-\bar{c}}^{2\alpha+(1-\lambda)} \left(\frac{\alpha}{2} - \frac{n-1+\lambda}{4}\right) dn + \frac{\partial \bar{c}}{\partial \lambda}(t-\bar{c})\right] \\
&= \delta\left[\frac{(\bar{c}+2\alpha+\frac{1}{2}-\lambda)^2}{8v} + \frac{\bar{c}+2\alpha+\frac{1}{2}-\lambda}{4v+\bar{c}+2\alpha+\frac{1}{2}-\lambda}(t-\bar{c})\right] \\
&= \frac{\delta(\bar{c}+2\alpha+\frac{1}{2}-\lambda)}{8v(4v+\bar{c}+2\alpha+\frac{1}{2}-\lambda)}\left[(\bar{c}+2\alpha+\frac{1}{2}-\lambda)(4v+\bar{c}+2\alpha+\frac{1}{2}-\lambda) + 8v(t-\bar{c})\right] \\
&= \frac{\delta(\bar{c}+2\alpha+\frac{1}{2}-\lambda)}{8v(4v+\bar{c}+2\alpha+\frac{1}{2}-\lambda)}\left[2\left(-6v\bar{c} + v + 4\alpha^2 + 4\alpha v - 2\lambda v\right)\right],
\end{aligned}$$

where the last equality is obtained by first substituting $\bar{c}^2$ with $-\Big[(4\alpha+1-2\lambda+8v)\bar{c} - 4\alpha^2 + \alpha(2-$

$4\lambda) - (1-\lambda)\lambda + 8tv + \frac{1}{4}\Big]$ (due to $F(\overline{c}) = 0$) and then simplifying the expression. Therefore,

$$\text{sign}\left(\frac{\partial \Pi}{\partial \lambda}\right) = \text{sign}\left(-6v\overline{c} + v + 4\alpha^2 + 4\alpha v - 2\lambda v\right).$$

Let

$$Q_1(v) \equiv 4\alpha^2 + v(4\alpha - 6\overline{c} - 2\lambda + 1), \tag{A8}$$

where $\overline{c}$ is also a function of v as given by equation (A7). Then,

$$\begin{aligned} Q_1'(v) &= -6\overline{c} - 6v\frac{\partial \overline{c}}{\partial v} + 4\alpha + 1 - 2\lambda \\ &= -6\overline{c} - 6v\left(-\frac{F_v}{F_{\overline{c}}}\right) + 4\alpha + 1 - 2\lambda \\ &= -6\overline{c} + \frac{48v\left(\overline{c} + t\right)}{8v + 2\overline{c} + 4\alpha + 1 - 2\lambda} + 4\alpha + 1 - 2\lambda. \end{aligned}$$

Let $R \equiv 4\alpha + 1 - 2\lambda$. Then, $R < 0$ by the assumption that $\lambda \in \left[\max\left\{\frac{1}{2} + 2\alpha, 1 - \alpha\right\}, 1\right]$ and $R + 2\overline{c} = 2(\overline{c} + 2\alpha + \frac{1}{2} - \lambda) > 0$ since $\overline{c} > -2\alpha - \frac{1}{2} + \lambda$. Therefore, we have

$$\begin{aligned} Q_1'(v) &= -6\overline{c} + \frac{48v\left(\overline{c} + t\right)}{8v + R + 2\overline{c}} + R \\ &= \frac{\left(R - 6\overline{c}\right)\left(R + 2\overline{c}\right) + 8v(R + 6t)}{8v + R + 2\overline{c}} < 0 \end{aligned}$$

when $t = 0$. That is, $Q_1(v)$ is decreasing in v when $t = 0$.

Also, if $v = \frac{\alpha^2}{\lambda - \frac{1}{2} - 2\alpha + t}$, we have $\overline{c} = \frac{1}{2} - (2\alpha + (1 - \lambda))$ and thus

$$Q_1\left(\frac{\alpha^2}{\lambda - \frac{1}{2} - 2\alpha + t}\right) = 4\alpha^2\left(\frac{4t}{-4\alpha + 2\lambda + 2t - 1} - 1\right) < 0$$

when $t = 0$.

Case 3. Condition: $1 - \lambda < \frac{1}{2} - \overline{c} \leq 2\alpha - (1 - \lambda)$.

Following a similar procedure as that of Case 2, we can derive that the condition for Case 3, $1 - \lambda < \frac{1}{2} - \overline{c} \leq 2\alpha - (1 - \lambda)$, can be rewritten as

$$\frac{(1-\lambda)(2\alpha - 1 + \lambda)}{2\lambda - 1 + 2t} < v \leq \frac{2\alpha^2 - (1-\lambda)^2}{3 - 4\alpha - 2\lambda + 2t}.$$

Furthermore, we can also derive

$$\text{sign}\left(\frac{\partial \Pi}{\partial \lambda}\right) = \text{sign}\left(Q_2(v)\right),$$

where

$$Q_2(v) \equiv 8\alpha^2 - 2v(1-4\alpha+4\overline{c}+2t) - 2(1-\lambda)^2, \tag{A9}$$

and $\overline{c}$ is the larger root of the following quadratic equation:

$$G(\overline{c}) \equiv \overline{c}^2 + (4\alpha - 1 + 4v)\overline{c} - 2\alpha + (1-\lambda)^2 + 4tv + \frac{1}{4} = 0, \tag{A10}$$

or

$$\overline{c} = \frac{1}{2}\left(-4\alpha + 1 - 4v + 2\sqrt{4\alpha^2 - \lambda^2 + 2\lambda - 4tv + 4v^2 + 8\alpha v - 2v - 1}\right). \tag{A11}$$

Note that $Q_1\left(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}\right) = Q_2\left(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}\right) = \frac{4\left(4\alpha^3-4\alpha(1-\lambda)^2+(2-\lambda)(1-\lambda)^2+\alpha^2(2t-1)\right)}{-4\alpha-2\lambda+2t+3}$ at the border between Cases 2 and 3 (i.e., when $v = \frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}$).

In addition, since $\frac{1}{2} - \overline{c} \le 2\alpha - (1-\lambda)$, we have $2\alpha + \overline{c} - \frac{1}{2} \ge 1 - \lambda \ge 0$ (which also implies $G_{\overline{c}} > 0$).

$$\begin{aligned}
Q_2'(v) &= -8\overline{c} - 8v\frac{\partial \overline{c}}{\partial v} - 2 + 8\alpha - 4t \\
&= -8\left(\overline{c} + v\left(-\frac{G_v}{G_{\overline{c}}}\right)\right) - 2(1-4\alpha) - 4t \\
&= -8\left(\overline{c} - \frac{2v(\overline{c}+t)}{2\alpha + \overline{c} + 2v - \frac{1}{2}}\right) - 2(1-4\alpha) - 4t \\
&= -8\left(\frac{2\alpha + \overline{c} - \frac{1}{2} - 2vt}{2\alpha + \overline{c} - \frac{1}{2} + 2v}\right) - 2(1-4\alpha) - 4t.
\end{aligned} \tag{A12}$$

Clearly, $Q_2'(v) < 0$ when $t = 0$. Thus, $Q_2(v)$ is decreasing in v when $t = 0$.

Also, if $v = \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$, we have $\overline{c} = \frac{1}{2} - (1-\lambda)$ and thus

$$Q_2\left(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}\right) = \frac{2(2\alpha-1+\lambda)(\alpha-(1-\lambda)\lambda+2\alpha t)}{2\lambda-1+2t} > 0$$

for any $t \ge 0$, since $\lambda \ge \frac{1}{2}$ and $\alpha > (1-\lambda)\lambda$ which is due to the assumption $\lambda \ge 1-\alpha$.

Case 4. Condition: $\frac{1}{2} - \overline{c} \le 1 - \lambda$.

In this case, $m(\frac{1}{2}-\overline{c}) = (\frac{1}{2}-\overline{c})\left(\alpha - \frac{1-\lambda}{2}\right)$, and equation (A2) is reduced to $\frac{1}{v}(\frac{1}{2}-\overline{c})\left(\alpha - \frac{1-\lambda}{2}\right) - \overline{c} = t$. Therefore,

$$\overline{c} = \frac{1}{2} - \frac{(2t+1)v}{2\alpha+\lambda-1+2v},$$

and

$$\underline{n} = \frac{(2t+1)v}{2\alpha+\lambda-1+2v}.$$

Note that $2\alpha + \lambda - 1 > \alpha + \lambda - 1 \ge 0$ by the assumption that $\lambda \ge 1 - \alpha$.

The condition for Case 4, $\frac{1}{2} - \bar{c} \leq 1 - \lambda$, can be reduced to

$$v \leq \frac{(1-\lambda)(2\alpha - 1 + \lambda)}{2\lambda - 1 + 2t}. \tag{A13}$$

By equation (A4), we have

$$\begin{aligned}
\frac{\partial \Pi}{\partial \lambda} &= \delta \left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda} dn + \frac{\partial \bar{c}}{\partial \lambda}(t - \bar{c}) \right] \\
&= \delta \Bigg[\frac{1}{v} \Bigg(\int_{\underline{n}}^{1-\lambda} \frac{\partial n \left(\alpha - \frac{1-\lambda}{2} \right)}{\partial \lambda} dn + \int_{1-\lambda}^{2\alpha + (1-\lambda)} \frac{\partial \frac{4\alpha^2 + 4\alpha(n-1+\lambda) - (n-1+\lambda)^2}{8}}{\partial \lambda} dn \\
&\quad + \int_{2\alpha+(1-\lambda)}^{\frac{1}{2}} \frac{\partial \alpha^2}{\partial \lambda} dn \Bigg) + \frac{\partial \bar{c}}{\partial \lambda}(t - \bar{c}) \Bigg] \\
&= \delta \left[-\frac{\underline{n}^2 - (1-\lambda)(4\alpha + \lambda - 1)}{4v} + \frac{\partial \bar{c}}{\partial \lambda}(t - \bar{c}) \right] \\
&= \delta \left[\frac{(1-\lambda)(4\alpha + \lambda - 1)}{4v} - v(2t+1) \frac{3(2\alpha + \lambda - 1) + 2v + 2t\,(2\alpha - 1 + \lambda - 2v)}{4(2\alpha - 1 + \lambda + 2v)^3} \right],
\end{aligned}$$

where the last equality is obtained by plugging in the expressions of $\bar{c}$ and $\underline{v}$ and simplifying the terms. Let $S_a(v) \equiv \frac{(1-\lambda)(4\alpha+\lambda-1)}{4v}$, $S_b(v) \equiv -v\frac{3(2\alpha+\lambda-1)+2v+2t(2\alpha-1+\lambda-2v)}{4(2\alpha-1+\lambda+2v)^3}$, and $S(v) \equiv S_a(v) + (2t + 1)S_b(v)$. Then, $\text{sign}(\frac{\partial \Pi}{\partial \lambda}) = \text{sign}(S(v))$.

We next show $S(v)$ is decreasing in v. First note that $S_a(v)$ is clearly decreasing in v since $4\alpha + \lambda - 1 > \alpha + \lambda - 1 \geq 0$. Let $X \equiv 2\alpha - 1 + \lambda > 0$, then $S_b(v) = -v\frac{3X+2v+2t(X-2v)}{4(X+2v)^3}$. Taking the first-order derivative of $S_b(v)$ w.r.t. v gives

$$\begin{aligned}
S_b'(v) &= \frac{(4-8t)v^2 + 8(2t+1)vX - (2t+3)X^2}{4(X+2v)^4} \\
&= \frac{4v^2 + 8vX - 3X^2 - 2t(4v^2 - 8vX + X^2)}{4(X+2v)^4}.
\end{aligned}$$

Let

$$\zeta(\lambda) \equiv \frac{1-\lambda}{2\lambda - 1 + 2t}.$$

Then, $\zeta(\lambda)$ is decreasing in λ, since $\zeta'(\lambda) = \frac{-(2t+1)}{(2\lambda-1+2t)^2} < 0$.

By the assumption that $\lambda \in \left[\max\left\{\frac{1}{2} + 2\alpha, 1 - \alpha\right\}, 1\right]$, we know $3\lambda \geq \frac{1}{2} + 2\alpha + 2(1-\alpha) = \frac{5}{2}$, and thus $\frac{5}{6} \leq \lambda \leq 1$. Therefore,

$$\zeta(\frac{5}{6}) \geq \zeta(\lambda) \geq \zeta(1).$$

In addition, by inequality (A13), we have

$$v \leq \zeta(\lambda)X.$$

Therefore,

$$\begin{aligned}
4v^2 + 8vX - 3X^2 &= 4(v+X)^2 - 7X^2 \\
&\leq 4\left(\zeta(\lambda)X + X\right)^2 - 7X^2 \\
&\leq 4\left(\zeta(1)X + X\right)^2 - 7X^2 \\
&= -3X^2. \qquad \text{(A14)}
\end{aligned}$$

Also, $4v^2 - 8vX + X^2 = 4(v-X)^2 - 3X^2$ is increasing in v when $v < X$. Furthermore, $v \leq \zeta(\lambda)X \leq \zeta(\frac{5}{6})X = \frac{\frac{1}{6}X}{\frac{2}{3}+2t} = \frac{X}{4+12t} < X$, so

$$\begin{aligned}
2t(4v^2 - 8vX + X^2) &= 2t\left[4(v-X)^2 - 3X^2\right] \\
&\geq 2t\left[4\left(\frac{X}{4+12t} - X\right)^2 - 3X^2\right] \\
&= \frac{3t\left(12t^2-1\right)}{2(3t+1)^2}X^2 \\
&\geq -3X^2, \qquad \text{(A15)}
\end{aligned}$$

where the last inequality is because

$$\begin{aligned}
\frac{t\left(12t^2-1\right)}{2(3t+1)^2} \geq -1 &\Leftarrow t\left(12t^2-1\right) + 2(3t+1)^2 \geq 0 \\
&\Leftarrow 12t^3 + 18t^2 + 11t + 2 \geq 0,
\end{aligned}$$

which is clearly true for any $t \geq 0$.

Inequality (A14) minus inequality (A15) gives

$$4v^2 + 8vX - 3X^2 - 2t(4v^2 - 8vX + X^2) \leq -3X^2 - (-3X^2) = 0.$$

and thus $S_b'(v) \leq 0$. Therefore, $S(v) = S_a(v) + S_b(v)$ is decreasing in v when $v \leq \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$ for any $t \geq 0$.

Furthermore,

$$S\left(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}\right) = \frac{(2\lambda-1+2t)(\alpha-(1-\lambda)\lambda+t(2\alpha-1+\lambda))}{(2t+1)(2\alpha-1+\lambda)} > 0,$$

since $\lambda \geq \frac{5}{6} > \frac{1}{2}$ and $\alpha > (1-\lambda)\lambda$. Thus, $S(v) > 0$ for all $v \leq \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$, which means $\text{sign}(\frac{\partial \Pi}{\partial \lambda}) = \text{sign}(S(v)) > 0$ when $v \leq \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$.

Summarizing the four cases above, we have now shown that for any $t \geq 0$,

- $\frac{\partial \Pi}{\partial \lambda} = 0$ when $v \geq \frac{\alpha^2}{\lambda - \frac{1}{2} - 2\alpha + t}$,
- $\frac{\partial \Pi}{\partial \lambda} > 0$ when $v \leq \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$, and
- $\text{sign}\left(\frac{\partial \Pi}{\partial \lambda}\right) = \text{sign}\left(Q(v)\right)$ when $\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t} < v < \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}$,

where

$$Q(v) \equiv \begin{cases} Q_1(v) & \text{if } \frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t} < v < \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}, \\ Q_2(v) & \text{if } \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t} < v \leq \frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}, \end{cases}$$

and $Q_1(v)$ and $Q_2(v)$ are defined by equations (A8) and (A9), respectively. In the proof for Case 2 and 3, we have shown that $Q_1\left(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}\right) = Q_2\left(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}\right)$, which means $Q(v)$ is continuous. We have also proved that when $t = 0$, which is the assumption in our baseline model, $Q_1(v)$ and $Q_2(v)$ are both decreasing in v, and thus $Q(v)$ is decreasing in v. Furthermore, we have shown that $Q(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}) = Q_2(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}) > 0$ for any $t \geq 0$, and $Q(\frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}) = Q_1(\frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}) < 0$ when $t = 0$. Therefore, in our baseline model where $t = 0$, we can conclude that there exist $\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t} < \underline{v} < \overline{v} = \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}$ such that $\text{sign}(\frac{\partial \Pi}{\partial \lambda}) = \text{sign}(Q(v)) < 0$ iff $v \in (\underline{v}, \overline{v})$. Let $\overline{\rho} = \frac{1}{\underline{v}}$ and $\underline{\rho} = \frac{1}{\overline{v}}$, then we know that $\frac{\partial \Pi}{\partial \lambda} < 0$ iff $\rho \in (\underline{\rho}, \overline{\rho})$.

□

Proof for Results Qualitatively Invariant to C. If we do not normalize C to 1/2 and let it be a free parameter, then the marketers are uniformly distributed on $[0, C]$ where the density of marketers at each c is equal to $\frac{1}{2C}$. Thus, the market clearing condition should be modified as follows:

$$\frac{1}{2} - \underline{n} = \overline{c} \cdot \left(\frac{1}{2C}\right). \tag{A16}$$

Let $\rho' \equiv \frac{\rho}{2C}$, $\overline{c}' \equiv \frac{\overline{c}}{2C}$, and $t' \equiv \frac{t}{2C}$. Then, $\frac{1}{2} - \underline{n} = \overline{c}'$, and thus $\frac{\partial \underline{n}}{\partial \lambda} = -\frac{\partial \overline{c}'}{\partial \lambda}$. Also, by equation (10), we have $M(\underline{n}) - \overline{c} = t$, which is equivalent to

$$M(\underline{n}) - \overline{c}' \cdot 2C = t' \cdot 2C. \tag{A17}$$

By equation (11), the platform's revenue can be re-written as:

$$\begin{aligned} \Pi &= \int_{\underline{n}}^{\frac{1}{2}} \delta\left(M(n) - \overline{c}\right) dn \\ &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} M(n) dn - (\frac{1}{2} - \underline{n})\overline{c}\right] \\ &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} M(n) dn - (\overline{c}')^2 \cdot 2C\right]. \end{aligned}$$

Therefore,

$$\begin{aligned}\frac{\partial \Pi}{\partial \lambda} &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda} dn - \frac{\partial \underline{n}}{\partial \lambda} M(\underline{n}) - 2\overline{c}' \frac{\partial \overline{c}'}{\partial \lambda} \cdot 2C\right] \\ &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \rho \cdot \frac{\partial m(n)}{\partial \lambda} dn + \frac{\partial \overline{c}'}{\partial \lambda}(M(\underline{n}) - 2\overline{c}' \cdot 2C)\right] \\ &= \delta\left[\int_{\underline{n}}^{\frac{1}{2}} \rho' \cdot 2C \cdot \frac{\partial m(n)}{\partial \lambda} dn + \frac{\partial \overline{c}'}{\partial \lambda}(2C(\overline{c}' + t') - 2\overline{c}' \cdot 2C)\right] \\ &= \delta \cdot 2C\left[\int_{\underline{n}}^{\frac{1}{2}} \rho' \cdot \frac{\partial m(n)}{\partial \lambda} dn + \frac{\partial \overline{c}'}{\partial \lambda}(t' - \overline{c}')\right], \end{aligned} \tag{A18}$$

where the second equality is due to $\overline{c}' = \frac{1}{2} - \underline{n}$, and the third equality is because of equation (A17). Thus,

$$\operatorname{sign}\left(\frac{\partial \Pi}{\partial \lambda}\right) = \operatorname{sign}\left[\int_{\underline{n}}^{\frac{1}{2}} \rho' \cdot \frac{\partial m(n)}{\partial \lambda} dn + \frac{\partial \overline{c}'}{\partial \lambda}(t' - \overline{c}')\right]. \tag{A19}$$

That is, the sign of $\frac{\partial \Pi}{\partial \lambda}$ expressed in equation (A19) resembles that in equation (A5) in our proof of Proposition 1, except that some parameters are rescaled. Thus, our key argument about the impact of matching accuracy on platform revenue is qualitatively invariant to the level of C.

$\square$

Proof of Proposition 2. $\gamma_0 > 0$ is equivalent to $t > 0$. In the proof of Proposition 1, we did not let $t = 0$ until the end, and we also specified which results were true only when $t = 0$. Here are the results in the proof of Proposition 1 that are still true when $t > 0$:

- $\frac{\partial \Pi}{\partial \lambda} = 0$ when $v \geq \frac{\alpha^2}{\lambda - \frac{1}{2} - 2\alpha + t}$.
- $\frac{\partial \Pi}{\partial \lambda} > 0$ when $v \leq \frac{(1-\lambda)(2\alpha - 1 + \lambda)}{2\lambda - 1 + 2t}$.
- $\operatorname{sign}\left(\frac{\partial \Pi}{\partial \lambda}\right) = \operatorname{sign}(Q(v))$ when $\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t} < v < \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}$, where

$$Q(v) \equiv \begin{cases} Q_1(v) & \text{if } \frac{2\alpha^2 - (1-\lambda)^2}{3 - 4\alpha - 2\lambda + 2t} < v < \frac{\alpha^2}{\lambda - \frac{1}{2} - 2\alpha + t}, \\ Q_2(v) & \text{if } \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda - 1 + 2t} < v \leq \frac{2\alpha^2 - (1-\lambda)^2}{3 - 4\alpha - 2\lambda + 2t}, \end{cases}$$

  and $Q_1(v)$ and $Q_2(v)$ are defined by equations (A8) and (A9), respectively.
- $Q(v)$ is continuous in v.
- $Q(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}) > 0$.

Here are the results in the proof of Proposition 1 that are not necessarily true when $t > 0$:

- $Q(v)$ is decreasing in v.

- $Q(\frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}) < 0$.

This means $Q(v)$ may not be negative at all for any $v \in \left(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}, \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}\right)$. Denote $v_1 \equiv \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}$ and $v_2 \equiv \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$. Proving the results stated in Proposition 2 is equivalent to showing that

- when $t < \bar{t}$, there exist $v_2 \leq \underline{v} < \overline{v} \leq v_1$ such that $Q(v) < 0$ iff $v \in (\underline{v}, \overline{v})$;

- when $t \geq \bar{t}$, $Q(v) \geq 0$ for any $v \in (v_2, v_1)$.

To prove this, we implement the following four steps.

Step 1. We show $Q'(v_2) \leq 0$. That is, $Q(\cdot)$ is initially decreasing.

To show $Q'(v_2) = Q_2'(v_2) \leq 0$, we plug in $v = \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$ and $c = \frac{1}{2} - (1-\lambda)$ to equation (A12), which gives

$$Q'(v_2) = \frac{6 + 8\alpha - 16\lambda - 8t(1 - 2\alpha + 2\lambda + t)}{2t+1}.$$

Since $\alpha \leq \frac{1}{4}$, we know that $1 - 2\alpha \geq \frac{1}{2} > 0$ so $8t(1 - 2\alpha + 2\lambda + t) \geq 0$. Also, since $\lambda \geq \frac{1}{2}$, we know that $6 + 8\alpha - 16\lambda \leq 6 + (8)(\frac{1}{4}) - (16)(\frac{1}{2}) = 0$. Therefore, $6 + 8\alpha - 16\lambda - 8t(1 - 2\alpha + 2\lambda + t) \leq 0$, and thus $Q'(v_2) \leq 0$.

Step 2. We show $Q_1''(v) \geq 0$ and $Q_2''(v) \geq 0$ for any $v \in \left(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}, \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}\right)$. That is, $Q_1(\cdot)$ and $Q_2(\cdot)$ are both convex functions.

Based on equations (A8) and (A7), we have

$$Q_1(v) = 4\left(6v^2 + \alpha^2 + v\left(4\alpha - 2\lambda + 1 - 3\sqrt{2\alpha^2 - 2tv + 4v^2 + 4\alpha v - 2\lambda v + v}\right)\right), \tag{A20}$$

and taking the second-order derivative w.r.t. v gives $Q_1''(v) = \Big[3\Big(1-3v\big(8\alpha-4\lambda+4(12\alpha^2-4\alpha\lambda+\lambda^2+t^2-t(4\alpha-2\lambda+1))\big)-(48v^2+8\alpha^2)(4\alpha-2\lambda-2t+1)-128v^3\Big)\Big]/\Big[\Big(2\alpha^2+v(4\alpha-2\lambda-2t+1)+4v^2\Big)^{3/2}\Big]+48.$

Based on equations (A9) and (A11), we have

$$Q_2(v) = 8v^2 + 4\alpha^2 - v\left(2t + 3 - 12\alpha + 4\sqrt{4\alpha^2 - (1-\lambda)^2 + 2v(2v - 2t - 1) + 8\alpha v}\right) - (1-\lambda)^2, \tag{A21}$$

and taking the second-order derivative w.r.t. v gives $Q_2''(v) = \Big[8\Big(2(1-4\alpha)(2\alpha-\lambda+1)(2\alpha+\lambda-1) - 12t^2v + 4t\big(4\alpha^2 - (\lambda-1)^2 + 12\alpha v + 3v(4v-1)\big) - 32v^3 + (24-96\alpha)v^2 + 3v(8\alpha(1-4\alpha) - 4(2-\lambda)\lambda + 3)\Big)\Big]/\Big[\Big(4\alpha^2 - (1-\lambda)^2 + v(8\alpha - 4t - 2) + 4v^2\Big)^{3/2}\Big] + 32.$

We show $Q_1''(v) \geq 0$ and $Q_2''(v) \geq 0$ for any $v \in \left(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}, \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}\right)$ using the `FindInstance` function in Mathematica.[23] An empty set is returned when we apply the `FindInstance` function to

[23] The documentation of the `FindInstance` function can be found at `https://reference.wolfram.com/language/ref/FindInstance.html`.

the condition $Q_1''(v) < 0$, together with the range of v ($\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t} < v < \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}$) and the other model assumptions ($0 \le \alpha \le \frac{1}{4} \wedge \max\left(2\alpha+\frac{1}{2}, 1-\alpha\right) \le \lambda \le 1 \wedge t \ge 0$). Similarly, `FindInstance` function also shows that the intersection set of $Q_2''(v) < 0$, $\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t} < v \le \frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}$, and the other model assumptions ($0 \le \alpha \le \frac{1}{4} \wedge \max\left(2\alpha+\frac{1}{2}, 1-\alpha\right) \le \lambda \le 1 \wedge t \ge 0$) is empty. This confirms that $Q_1''(v) < 0$ or $Q_2''(v) < 0$ is impossible for any $v \in \left(\frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}, \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}\right)$, which means that we have $Q_1''(v) \ge 0$ and $Q_2''(v) \ge 0$.

Step 3. We show that at the border between $Q_1(v)$ and $Q_2(v)$, i.e., at $v = \frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}$, we have $Q_2'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) > 0 \Rightarrow Q_1'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) \ge 0$. This, together with the results obtained in Steps 1 and 2, ensures that $Q(v)$ is either always decreasing in v or first decreasing and then increasing in v when $v \in (v_1, v_2)$. Furthermore, this result implies that the set $\{v \in (v_1, v_2) : Q(v) < 0\}$, if non-empty, can only be a continuous range of v, i.e., $(\underline{v}, \overline{v})$ where $v_2 \le \underline{v} < \overline{v} \le v_1$.

Showing $Q_2'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) > 0 \Rightarrow Q_1'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) \ge 0$ is equivalent to showing it is impossible that both $Q_2'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) > 0$ and $Q_1'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) < 0$ hold simultaneously.

Based on equations (A20) and (A21), we can take the first-order derivative w.r.t. v to get $Q_1'(v) = 16\alpha - 8\lambda - \frac{6v(4\alpha-2\lambda-2t+8v+1)}{\sqrt{2\alpha^2+v(4\alpha-2\lambda-2t+1)+4v^2}} - 12\sqrt{2\alpha^2+v(4\alpha-2\lambda-2t+1)+4v^2} + 48v + 4$, and $Q_2'(v) = 24\alpha - 8\sqrt{4\alpha^2-(1-\lambda)^2+v(8\alpha-4t-2)+4v^2} + \frac{8v(1-4\alpha+2t-4v)}{\sqrt{4\alpha^2-(1-\lambda)^2+v(8\alpha-4t-2)+4v^2}} - 4t + 32v - 6$. Mathematica's `FindInstance` function confirms that the interaction set of $Q_2'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) > 0$ and $Q_1'(\frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}) < 0$, together with the model assumptions ($0 \le \alpha \le \frac{1}{4} \wedge \max\left(2\alpha+\frac{1}{2}, 1-\alpha\right) \le \lambda \le 1 \wedge t \ge 0$), is empty.

Step 4. We finish the proof by analyzing the effect of t on the shape of $Q(v)$. We rewrite $Q(v)$, $Q_1(v)$, and $Q_2(v)$ all as functions of both v and t, i.e., we let $Q(v,t) = Q(v)$, $Q_1(v,t) = Q_1(v)$, and $Q_2(v,t) = Q_2(v)$.

Let $v_0(t) \equiv \frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}$. Then $Q(v,t) = Q_1(v,t)$ if $v < v_0(t)$ and $Q(v,t) = Q_2(v,t)$ if $v \ge v_0(t)$. Also denote $v_1(t) \equiv \frac{\alpha^2}{\lambda-\frac{1}{2}-2\alpha+t}$ and $v_2(t) \equiv \frac{(1-\lambda)(2\alpha-1+\lambda)}{2\lambda-1+2t}$. Clearly, $v_0(t)$, $v_1(t)$, and $v_2(t)$ are all decreasing in t.

Step 4.1. We show there exists $T > 0$ s.t. $Q(v,t) > 0$ for any $v \in (v_2(t), v_1(t))$ if $t > T$.

Based on equation (A20), we know that $Q_1(v,t) > 0$ is equivalent to

$$6v^2 + \alpha^2 + v\left(4\alpha - 2\lambda + 1\right) > 3v\sqrt{2\alpha^2 - 2tv + 4v^2 + 4\alpha v - 2\lambda v + v},$$

or

$$q_1(v,t) \equiv [6v^2 + \alpha^2 + v\left(4\alpha - 2\lambda + 1\right)]^2 - 3v(2\alpha^2 - 2tv + 4v^2 + 4\alpha v - 2\lambda v + v) > 0.$$

Simplifying $q_1(v,t)$ gives

$$q_1(v,t) = 3v^2[v(6t + 1 + 4\alpha - 2\lambda) - 2\alpha^2] + [v(1 + 4\alpha - 2\lambda) + \alpha^2]^2.$$

Similarly, we can derive from equation (A21) that $Q_2(v,t) > 0$ is equivalent to $q_2(v,t) > 0$, where

$$q_2(v,t) = 16v^3[2t - 1 + 4\alpha] + [v(3 + 2t - 12\alpha) - 4\alpha^2 + (1-\lambda)^2]^2.$$

It is straightforward to see that if

$$t > \frac{1-4\alpha}{2}, \tag{A22}$$

then $2t - 1 + 4\alpha > 0$ and thus $q_2(v,t) > 0$.

When $t > \frac{1-4\alpha}{2}$, we have $6t + 1 + 4\alpha - 2\lambda > 6\frac{1-4\alpha}{2} + 1 + 4\alpha - 2\lambda = 4 - 8\alpha - 2\lambda \geq 4 - 8\alpha - 2 = 2(1-4\alpha) \geq 0$. Also, since $Q(v,t) = Q_1(v,t)$ if $v < v_0(t) \equiv \frac{2\alpha^2-(1-\lambda)^2}{3-4\alpha-2\lambda+2t}$, we know that

$$\begin{aligned}
q_1(v,t) &= 3v^2[v(6t + 1 + 4\alpha - 2\lambda) - 2\alpha^2] + [v(1 + 4\alpha - 2\lambda) + \alpha^2]^2 \\
&> 3v^2\left[\frac{2\alpha^2 - (1-\lambda)^2}{3 - 4\alpha - 2\lambda + 2t}(6t + 1 + 4\alpha - 2\lambda) - 2\alpha^2\right].
\end{aligned}$$

Consider the case when

$$t > \frac{5-12\alpha}{2}, \tag{A23}$$

and thus $t > \frac{5-12\alpha-2\lambda}{2}$, which further implies $(6t + 1 + 4\alpha - 2\lambda) > 2(3 - 4\alpha - 2\lambda + 2t)$. Therefore,

$$\begin{aligned}
q_1(v,t) &> 3v^2\left[\frac{2\alpha^2 - (1-\lambda)^2}{3 - 4\alpha - 2\lambda + 2t}(6t + 1 + 4\alpha - 2\lambda) - 2\alpha^2\right] \\
&> 3v^2\left[2(2\alpha^2 - (1-\lambda)^2) - 2\alpha^2\right] \\
&= 6v^2[\alpha^2 - (1-\lambda)^2] \geq 0,
\end{aligned}$$

by the assumption that $\lambda \geq 1 - \alpha$.

In summary, we have shown that $q_1(v,t) > 0$ if $t > \frac{5-12\alpha}{2}$ and $q_2(v,t) > 0$ if $t > \frac{1-4\alpha}{2}$. Note that $\frac{5-12\alpha}{2} - \frac{1-4\alpha}{2} = \frac{4(1-2\alpha)}{2} > 0$. Thus, if we let $T = \frac{5-12\alpha}{2}$, we know that $q_1(v,t) > 0$ and $q_2(v,t) > 0$ if $t > T$.

Step 4.2. Consider $0 \leq t_1 < t_2 \leq T$. Then $v_0(t_1) > v_0(t_2)$. To finish the proof of Proposition 2, we only need to show the following two conditions cannot hold simultaneously:

$$\forall v \in (v_2(t_1), v_1(t_1)), Q(v,t_1) \geq 0, \tag{A24}$$

and

$$\exists v \in (v_2(t_2), v_1(t_2)), Q(v,t_2) < 0. \tag{A25}$$

We prove this by contradition.

First, consider any $v \geq v_0(t_1)$. In this case, $Q(v,t_1) = Q_1(v,t_1)$ and $Q(v,t_2) = Q_1(v,t_2)$. Note that for any $v > 0$, we can show that $Q_1(v,t)$ is increasing in t. This is because we can derive from equations (A8) and (A6) that

$$\frac{\partial Q_1}{\partial t} = -6v\frac{\partial \overline{c}}{\partial t} = 6v\frac{F_t}{F_{\overline{c}}} = 6v\frac{8v}{F_{\overline{c}}} > 0,$$

and we have already shown that $F_{\overline{c}} = -2\alpha - \frac{1}{2} + \lambda > 0$ in the proof of Proposition 1. Therefore, $Q_1(v,t_1) \leq Q_1(v,t_2)$, and thus $Q(v,t_2) < 0$ implies $Q(v,t_1) < 0$, so it cannot be the case that $Q(v,t_1) \geq 0$ and $Q(v,t_2) < 0$ when $v \geq v_0(t_1)$. So we only need to focus on $v < v_0(t_1)$.

We have already proved that $Q_2(v,t)$ is convex in v when $v < v_0(t)$, which means there are at most have two solutions of v to equation $Q_2(v,t) = 0$ for $v < v_0(t)$. This also implies there are at most two solutions of v to $q_2(v,t) = 0$ when $v < v_0(t)$. In addition, $q_2(v,t)$ is a cubic function. Furthermore, Mathematica's `FindInstance` function confirms that the interaction set of $\frac{\partial q_2(v,t)}{\partial v} \geq 0$, $q_2(v,t) = 0$, and $v < v_0(t)$, together with the model assumptions ($0 \leq \alpha \leq \frac{1}{4} \wedge \max\left(2\alpha + \frac{1}{2}, 1-\alpha\right) \leq \lambda \leq 1 \wedge t \geq 0$), is empty. That means $\frac{\partial q_2(v,t)}{\partial v} < 0$ whenever $q_2(v,t) = 0$, which further implies that there is *at most* one solution of v to equation $q_2(v,t) = 0$ when $v < v_0(t)$ (otherwise there must be a root of v such that $\frac{\partial q_2(v,t)}{\partial v} > 0$ according to the shape of a cubic function). Therefore, there is also *at most* one solution of v to equation $Q_2(v,t) = 0$ when $v < v_0(t)$.

Case 1. There is exactly one solution of $v = v_s$ to equation $Q_2(v,t_2) = 0$ when $v < v_0(t_2)$. Since we also have $Q_2(v_2(t),t) > 0$ and $\frac{\partial Q_2}{\partial v}(v_2(t),t) \leq 0$ from previous steps, we know that $Q_2 > 0$ if $v < v_s$ and $Q_2 < 0$ if $v > v_s$. Therefore, $Q_2(v_0(t_2),t_2) < 0$. Also, $Q_1(v_0(t_1),t_1) \geq 0$ by condition (A24). Mathematica's `FindInstance` function confirms that the intersection of $Q_2(v_0(t_2),t_2) < 0$, $Q_1(v_0(t_1),t_1) \geq 0$, and $0 \leq t_1 < t_2$, together with the model assumptions, is empty. Contradiction.

Case 2. There is no solution of v to equation $Q_2(v,t_2) = 0$ when $v < v_0(t_2)$. Then $Q(v,t_2) > 0$ for any $v < v_0(t_2)$, and $Q(v_0(t_2),t_2) = Q_1(v_0(t_2),t_2) = Q_2(v_0(t_2),t_2) > 0$. Thus, we only need to focus on $v \in (v_0(t_2), v_0(t_1))$. In the following, we discuss the sign of $\frac{\partial Q_1}{\partial v}(v_0(t_1),t_1)$ to complete the proof.

Case 2.1. $\frac{\partial Q_1}{\partial v}(v_0(t_1),t_1) > 0$. In this case, Mathematica's `FindInstance` function confirms that the intersection of $\frac{\partial Q_1}{\partial v}(v_0(t_1),t_1) > 0$, $\frac{\partial Q_1}{\partial v}(v_0(t_2),t_2) < 0$, and $0 \leq t_1 < t_2$, together with the model assumptions, is empty. Since we have proved the convexity of Q_2 in v, we know that $\frac{\partial Q_1}{\partial v} \geq 0$ for any $v > v_0(t_2)$. Therefore, for $v \in (v_0(t_2), v_0(t_1))$, we have $Q(v,t_2) = Q_1(v,t_2) > 0$. That means, we cannot find v such that $Q(v,t_2) < 0$. Contradiction.

Case 2.2. $\frac{\partial Q_1}{\partial v}(v_0(t_1),t_1) \leq 0$. In this case, by the convexity of Q_1 w.r.t. v, we know that $\frac{\partial Q_1}{\partial v} \leq 0$ for any $v \in (v_0(t_2), v_0(t_1))$. Thus, $Q(v,t_2) = Q_1(v,t_2) \geq Q_1(v,t_1) \geq Q_1(v_0(t_1),t_1) \geq 0$ for any $v \in (v_0(t_2), v_0(t_1))$. That means, we cannot find v such that $Q(v,t_2) < 0$. Contradiction.

□

Proof of Proposition 3. By equation (11), we have

$$\begin{aligned}\Pi^* &= \int_{\underline{n}}^{\frac{1}{2}} \delta^*(\lambda)\left(M(n)-\overline{c}\right)dn \\ &= \delta^*(\lambda)\left[\int_{\underline{n}}^{\frac{1}{2}} M(n)dn - (\frac{1}{2}-\underline{n})\overline{c}\right] \\ &= \delta^*(\lambda)\left[\int_{\underline{n}}^{\frac{1}{2}} M(n)dn - \overline{c}^2\right],\end{aligned}$$

where $\delta^*(\lambda)$ is the optimal commission rate chosen by the platform when the matching accuracy is λ. Therefore, by the envelope theorem, we have

$$\begin{aligned}\frac{\partial \Pi^*}{\partial \lambda} = \frac{\partial \Pi}{\partial \lambda}|_{\delta=\delta^*(\lambda)} &= \delta^*(\lambda)\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda}dn - \frac{\partial \underline{n}}{\partial \lambda}M(\underline{n}) - 2\overline{c}\frac{\partial \overline{c}}{\partial \lambda}\right] \\ &= \delta^*(\lambda)\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda}dn + \frac{\partial \overline{c}}{\partial \lambda}(M(\underline{n}) - 2\overline{c})\right] \\ &= \delta^*(\lambda)\left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial M(n)}{\partial \lambda}dn + \frac{\partial \overline{c}}{\partial \lambda}\left(\frac{\gamma_0}{1-\delta^*(\lambda)} - \overline{c}\right)\right],\end{aligned}$$

where the second equality is due to $\overline{c} = \frac{1}{2} - \underline{n}$, and the third equality is because $M(\underline{n}) - \overline{c} = p^*(\underline{n}) = \frac{\gamma_0}{1-\delta}$ by Lemma 2. Since $\frac{\partial M(n)}{\partial \lambda} \geq 0$, and we have shown in the proof of Proposition 1 by IFT that $\frac{\partial \overline{c}}{\partial \lambda} \geq 0$, to complete the proof we only need to show

$$\frac{\gamma_0}{1-\delta^*(\lambda)} - \overline{c} \geq 0.$$

To see this, consider the first-order condition for the platform solves for the optimal $\delta^*(\lambda)$:

$$\begin{aligned}0 &= \frac{\partial \Pi}{\partial \delta}|_{\delta=\delta^*(\lambda)} \\ &= \int_{\underline{n}}^{\frac{1}{2}} M(n)dn - \overline{c}^2 + \delta^*(\lambda)\left[-\frac{\partial \underline{n}}{\partial \delta}|_{\delta=\delta^*(\lambda)}M(\underline{n}) - 2\overline{c}\frac{\partial \overline{c}}{\partial \delta}|_{\delta=\delta^*(\lambda)}\right] \\ &= \frac{\Pi^*}{\delta^*(\lambda)} + \delta^*(\lambda)\left[\frac{\partial \overline{c}}{\partial \delta}|_{\delta=\delta^*(\lambda)}\left(\overline{c} + \frac{\gamma_0}{1-\delta^*(\lambda)}\right) - 2\overline{c}\frac{\partial \overline{c}}{\partial \delta}|_{\delta=\delta^*(\lambda)}\right] \\ &= \frac{\Pi^*}{\delta^*(\lambda)} + \delta^*(\lambda)\left[\frac{\partial \overline{c}}{\partial \delta}|_{\delta=\delta^*(\lambda)}\left(\frac{\gamma_0}{1-\delta^*(\lambda)} - \overline{c}\right)\right].\end{aligned}$$

Thus, we have

$$\frac{\gamma_0}{1-\delta^*(\lambda)} - \overline{c} = \frac{-\Pi^*}{\delta^*(\lambda)^2 \frac{\partial \overline{c}}{\partial \delta}|_{\delta=\delta^*(\lambda)}}$$

Clearly, $\frac{\Pi^*}{\delta^*(\lambda)^2} \geq 0$. By equations (9) and (10), we know that $\overline{c}$ is determined by solving the following equation:

$$M(\frac{1}{2}-\bar{c})-\bar{c}=\frac{\gamma_0}{1-\delta},$$

where $M(\cdot)$ is an increasing function. Thus, LHS is decreasing in $\bar{c}$. Also, as δ increases, RHS is increasing. Therefore, $\bar{c}$ decreases in δ, i.e., $\frac{\partial \bar{c}}{\partial \delta}<0$ and in particular, $\frac{\partial \bar{c}}{\partial \delta}|_{\delta=\delta^*(\lambda)}<0$. Therefore, we have

$$\frac{\gamma_0}{1-\delta^*(\lambda)}-\bar{c}=\frac{-\Pi^*}{\delta^*(\lambda)^2\frac{\partial \bar{c}}{\partial \delta}|_{\delta=\delta^*(\lambda)}}\geq 0.$$

□

Proof of Corollary 1. From the proof of Proposition 3 we know that

$$\frac{\partial \Pi}{\partial \lambda}=\delta\left[\int_{\underline{n}}^{\frac{1}{2}}\frac{\partial M(n)}{\partial \lambda}dn+\frac{\partial \bar{c}}{\partial \lambda}\left(\frac{\gamma_0}{1-\delta}-\bar{c}\right)\right],$$

and that

$$\frac{\partial \Pi}{\partial \delta}=\frac{\Pi}{\delta}+\delta\left[\frac{\partial \bar{c}}{\partial \delta}\left(\frac{\gamma_0}{1-\delta}-\bar{c}\right)\right].$$

If $\frac{\partial \Pi}{\partial \lambda}<0$, we know it must be the case that $\frac{\gamma_0}{1-\delta}-\bar{c}<0$ since $\frac{\partial M(n)}{\partial \lambda}\geq 0$. Therefore, $\frac{\partial \Pi}{\partial \delta}=\frac{\Pi}{\delta}+\delta\left[\frac{\partial \bar{c}}{\partial \delta}\left(\frac{\gamma_0}{1-\delta}-\bar{c}\right)\right]>0$, since $\frac{\partial \bar{c}}{\partial \delta}<0$.

□

Proof of Lemma 3. The proof of Lemma 3 is similar to that of Lemma 2. In equilibrium, the indifference condition still holds, and there still exists a marginal marketer $\bar{c}$ such that the participating marketers are $\mathcal{M}=[0,\bar{c}]$. The equilibrium price is still $p^*(n)=M(n)-\bar{c}$. The only difference is the influencers' incentive. Unlike in the proof of Lemma 2, now an influencer's individual rationality implies

$$V(p^*(n))=(1-\delta)p^*(n)\geq \gamma_0+\gamma_1\rho n,$$

or

$$(1-\delta)[M(n)-\bar{c}]\geq \gamma_0+\gamma_1\rho n.$$

Note that $M(n)=\rho m(n)$, so this condition is equivalent to

$$W(n):=m(n)-\frac{\gamma_1}{1-\delta}n\geq\frac{\frac{\gamma_0}{1-\delta}+\bar{c}}{\rho}.$$

Note that

$$\frac{dW}{dn}|_{n=0}=m'(0)-\frac{\gamma_1}{1-\delta}=(\alpha-\frac{1-\lambda}{2})-\frac{\gamma_1}{1-\delta}>0$$

based on the assumption that $\gamma_1 < (1-\delta)(\alpha - \frac{1-\lambda}{2})$. Also,

$$\frac{dW}{dn}|_{n=\frac{1}{2}} = m'(\frac{1}{2}) - \frac{\gamma_1}{1-\delta} = 0 - \frac{\gamma_1}{1-\delta} < 0.$$

Therefore, $W(n)$ is first increasing and then decreasing in n, and thus the set of n such that $W(n) \geq \frac{\frac{\gamma_0}{1-\delta}+\bar{c}}{\rho}$ must be $[\underline{n}, \overline{n}]$, where $0 \leq \underline{n} \leq \overline{n} \leq \frac{1}{2}$. That means the influencers who participate must be those with $n \in \mathcal{N} = [\underline{n}, \overline{n}]$.

□

Proof of Proposition 4. Intuitively, if AI improvement hurts platform's revenue even when the platform can optimally adjust δ, it must hurt the platform when it cannot do so. Hence we only need to show the case when the commission rate δ is endogenously chosen by the platform, since the case with exogenous δ is nested.

We have shown in the proof of Lemma 3 that $W(n) = m(n) - \frac{\gamma_1}{1-\delta}n$ is first increasing and then decreasing in n. Suppose the function $W(n)$ peaks at $n = \hat{n}$. We can find $\hat{n}$ by setting $W'(\hat{n}) = 0$. Solving $W'(\hat{n}) = 0$ for $\hat{n}$ gives

$$\hat{n} = \begin{cases} 2\alpha - \frac{2\gamma}{1-\delta} & \text{if } \frac{2\gamma}{1-\delta} \geq 1-\lambda, \\ 2\alpha - \frac{4\gamma}{1-\delta} - \lambda + 1 & \text{if } \frac{2\gamma}{1-\delta} < 1-\lambda, \end{cases}$$

or equivalently,

$$\hat{n} = \max\left(\frac{2\gamma}{1-\delta}, 1-\lambda\right) + 2\alpha - \frac{4\gamma}{1-\delta}.$$

By Lemma 3, we know that the participating marketers are $[0, \bar{c}]$ and the participating influencers are $[\underline{n}, \overline{n}]$. We must have $\underline{n} < \hat{n} < \overline{n}$.

For any given δ, the equilibrium $(\bar{c}, \underline{n}, \overline{n})$ can be found by solving the following system of equations:

$$\begin{cases} \bar{c} = \overline{n} - \underline{n}, \\ W(\underline{n}) = \frac{\frac{\gamma_0}{1-\delta}+\bar{c}}{\rho}, \text{ where } \underline{n} < \hat{n}, \\ W(\overline{n}_1) = \frac{\frac{\gamma_0}{1-\delta}+\bar{c}}{\rho}, \text{ where } \overline{n}_1 > \hat{n}, \\ \overline{n} = \min\left(\overline{n}_1, \frac{1}{2}\right). \end{cases} \tag{A26}$$

The reason why we have $\overline{n}_1$ in the equations is that we have an upper bound $(\frac{1}{2})$ for n, so $\overline{n} = \frac{1}{2}$ if the larger solution to $W(n) = \frac{\frac{\gamma_0}{1-\delta}+\bar{c}}{\rho}$ is greater than $\frac{1}{2}$.

The equation system (A26) can be reduced to

$$\begin{cases} m(\underline{n}) - \frac{\gamma_1}{1-\delta}\underline{n} = \frac{\frac{\gamma_0}{1-\delta}+\min\left(\overline{n}_1, \frac{1}{2}\right)-\underline{n}}{\rho}, \text{ where } \underline{n} < \hat{n}, \\ m(\overline{n}_1) - \frac{\gamma_1}{1-\delta}\overline{n}_1 = \frac{\frac{\gamma_0}{1-\delta}+\min\left(\overline{n}_1, \frac{1}{2}\right)-\underline{n}}{\rho}, \text{ where } \overline{n}_1 > \hat{n}. \end{cases} \tag{A27}$$

After solving the equation system (A27) for any δ, the platform needs to maximize its revenue by choosing δ. Mathematically, the platform solves

$$\begin{aligned}\max_{\delta}\Bigg[\Pi &= \int_{\underline{n}}^{\overline{n}} \delta\left(M(n)-\overline{c}\right) dn \\ &= \int_{\underline{n}}^{\overline{n}} \delta\left(M(n) - \left(\min\left(\overline{n}_1, \frac{1}{2}\right) - \underline{n}\right)\right) dn\Bigg].\end{aligned} \tag{A28}$$

The equation system (A27) and the maximization problem (A28) are not analytically tractable. Since the proposition only speaks to existence, we rely on a numerical example to show the possibility of $\Pi^*(\lambda_1) > \Pi^*(\lambda_2)$, where Π^* represents the platform's optimal revenue with endogenous δ, and $\lambda_1 < \lambda_2$. Consider $\alpha = 0.15$, $\rho = 10$, $\gamma_0 = 0$, and $\gamma_1 = 0.001$. Numerically solving (A27) and (A28) using Mathematica's `FindRoot` and `FindMaximum` functions under different values of λ gives Table A1, which confirms the existence of such possibility.

Table A1: Equilibrium with $\alpha = 0.15, \rho = 10, \gamma_0 = 0, \gamma_1 = 0.001$

AI precision (λ)	Optimal platform revenue (Π^*)
0.85	0.010469
0.90	0.010018
0.95	0.010006
1.00	0.010357

□

Proof of Proposition 5. Without loss of generality, we demonstrate the proof by showing the case of $\alpha \in (\frac{1}{6}, \frac{1}{4})$ as is illustrated in Figure 4 on page 12. Note that, $M(n)$ corresponds to the regions 1, 4, 5 and 6 (or 1, 4 and 5) given by equation (5) when $1-\lambda \in (0, \alpha)$, whereas it corresponds to the regions 1, 2, and 5 when $1-\lambda \in (\alpha, 2\alpha)$. We consider the following three cases.

Case (i) $\lambda > 1 - \alpha$.

Given that $\rho n = N > 0$, $\forall n \in (0, \frac{1}{2})$, we have

$$M(n) = \begin{cases} N\left(\alpha - \frac{1-\lambda}{2}\right) & \text{if } n \leq 1-\lambda \\ \alpha N - \frac{N\left(n^2+(1-\lambda)^2\right)}{4n} & \text{if } n \in (1-\lambda, 2\alpha-1+\lambda) \\ \frac{N\left(4\alpha^2+4\alpha(n-1+\lambda)-(n-1+\lambda)^2\right)}{8n} & \text{if } n \in \left(2\alpha-1+\lambda, \min\left\{\frac{1}{2}, 2\alpha+1-\lambda\right\}\right) \\ \frac{N}{n}\alpha^2 & \text{if } n \in \left(2\alpha+1-\lambda, \frac{1}{2}\right) \end{cases}$$

We can show that $\frac{\partial M(n)}{\partial n} \leq 0, \forall n \in (0, \frac{1}{2})$. That is, the expected revenue from collaborating with an influencer declines with her content broadness. In this case, all influencers with $n \in [0, \overline{n}]$ participate in equilibrium. The market clearing condition implies $\overline{c} = \overline{n}$. The price of the marginal influencer is given by $p^*(\overline{n}) = M(\overline{n}) - \overline{c} = M(\overline{n}) - \overline{n} = 0$. In other words, we have

$$M(\overline{n}) = \overline{n}.$$

Taking derivatives on both sides w.r.t. λ, we get

$$\frac{\partial \overline{n}}{\partial \lambda} = \frac{\partial M(\overline{n})}{\partial \lambda} + \frac{\partial M(\overline{n})}{\partial n} \cdot \frac{\partial \overline{n}}{\partial \lambda}.$$

We can re-write the above equation as follows:

$$\frac{\partial M(\overline{n})}{\partial \lambda} = \left(1 - \frac{\partial M(\overline{n})}{\partial n}\right) \cdot \frac{\partial \overline{n}}{\partial \lambda} = \left(1 - \frac{\partial M(\overline{n})}{\partial n}\right) \cdot \frac{\partial \overline{c}}{\partial \lambda}.$$

We can verify that $\frac{\partial M(n)}{\partial \lambda \partial n} \leq 0$ given the expression of $M(n)$, and thus for any $n \leq \overline{n}$, we have

$$\frac{\partial p^*(n)}{\partial \lambda} = \frac{\partial M(n)}{\partial \lambda} - \frac{\partial \overline{c}}{\partial \lambda} > \frac{\partial M(\overline{n})}{\partial \lambda} - \frac{\partial \overline{c}}{\partial \lambda} = -\frac{\partial M(\overline{n})}{\partial n} \cdot \frac{\partial \overline{c}}{\partial \lambda} \geq 0.$$

The last inequality follows the fact that $\frac{\partial M(\overline{n})}{\partial n} \leq 0$. Since all influencers' equilibrium prices increase with λ, we must have $\frac{\partial \Pi}{\partial \lambda} > 0$.

Case (ii) $1 - \left(\frac{2+2\sqrt{2}+N}{3+2\sqrt{2}+\frac{N}{2}}\right)\alpha < \lambda \leq 1 - \alpha$.

In this case, $M(\cdot)$ corresponds to the regions 1, 2, and 5 given by equation (5) and thus we have

$$M(n) = \rho m(n) = \begin{cases} N\left(\alpha - \frac{1-\lambda}{2}\right) & \text{if } n \leq 2\alpha - (1-\lambda) \\ \frac{N}{8n}(n - 1 + \lambda + 2\alpha)^2 & \text{if } n \in (2\alpha - (1-\lambda), 1-\lambda) \\ \frac{N}{8n}\left(4\alpha^2 + 4\alpha(n-1+\lambda) - (n-1+\lambda)^2\right) & \text{if } n \in \left[1-\lambda, \frac{1}{2}\right) \end{cases}.$$

It is easy to see that $M(\cdot)$ remains constant if $n \leq 2\alpha - (1-\lambda)$, then strictly increases in $n \in \left(2\alpha - (1-\lambda), \sqrt{(2\alpha + (1-\lambda))^2 - 8\alpha^2}\right)$ and strictly decreases in $n \in \left[\sqrt{(2\alpha + (1-\lambda))^2 - 8\alpha^2}, \frac{1}{2}\right)$. Let $n = n_2$ denote the solution to $\frac{N}{8n}\left(4\alpha^2 + 4\alpha(n-1+\lambda) - (n-1+\lambda)^2\right) = N\left(\alpha - \frac{1-\lambda}{2}\right)$. Solving this equation yields $n_2 = \left(3 + 2\sqrt{2}\right)(1-\lambda) - \left(2 + 2\sqrt{2}\right)\alpha$. We can show that under the condition that $\lambda > 1 - \left(\frac{2+2\sqrt{2}+N}{3+2\sqrt{2}+\frac{N}{2}}\right)\alpha$, we must have $N\left(\alpha - \frac{1-\lambda}{2}\right) \geq n_2$.[24] That is, all influencers with $n \in [0, \overline{n}]$ participate in equilibrium, where $\overline{n} > n_2$. In other words, even though $M(\cdot)$ no longer monotonically decreases in n, the most niche influencers ($n \in (0, 2\alpha - (1-\lambda)]$) all still participate in equilibrium. Note that we still have $\frac{\partial M(\overline{n})}{\partial n} < 0$ in this case. Hence the same proof for case (i) applies to case (ii) as well and we must have $\frac{\partial \Pi}{\partial \lambda} > 0$.

[24]In fact, that is how we obtained this threshold $1 - \left(\frac{2+2\sqrt{2}+N}{3+2\sqrt{2}+\frac{N}{2}}\right)\alpha$.

Case (iii) $1-\frac{\left(N+4+2\sqrt{2}\right)\alpha}{\frac{N}{2}+4+2\sqrt{2}}<\lambda<\min\{1-\alpha,1-\left(\frac{2+2\sqrt{2}+N}{3+2\sqrt{2}+\frac{N}{2}}\right)\alpha\}$.

We can show that in this case, we must have $n_2-[2\alpha-(1-\lambda)]<N\left(\alpha-\frac{1-\lambda}{2}\right)<n_2$. The shape of $M(\cdot)$ in case (iii) is identical to case (ii). However, in this case, the most niche influencers ($n\in[0,2\alpha-(1-\lambda)]$) are all indifferent between participating or not. When influencers are indifferent between participating or not, we impose the following tie-breaking rule: among the influencers who are indifferent, only a fraction of them participate until market clears where all marketers who strictly prefer to participate are matched with an influencer. In this equilibrium, all influencers with $n\in(2\alpha-(1-\lambda),n_2)$ participate at price $p^*(n)=M(n)-\overline{c}$, where $\overline{c}=N\left(\alpha-\frac{1-\lambda}{2}\right)$. Additionally, a fraction of influencers with $n\in[0,2\alpha-(1-\lambda)]$ participate at price 0 and their mass is equal to $\overline{c}-(n_2-[2\alpha-(1-\lambda)])$ so market clears. The platform obtains a revenue of Π given by

$$\Pi=\delta\left[\int_{2\alpha-(1-\lambda)}^{1-\lambda}p^*(n)dn+\int_{1-\lambda}^{n_2}p^*(n)dn\right].$$

Taking the first-order derivative w.r.t. λ, we have

$$\frac{\partial\Pi}{\partial\lambda}=\delta\left[\int_{2\alpha-(1-\lambda)}^{n_2}\frac{\partial p^*(n)}{\partial\lambda}dn+\frac{\partial n_2}{\partial\lambda}\cdot\underbrace{p^*\left(n_2\right)}_{=0}-\underbrace{p^*\left(2\alpha-(1-\lambda)\right)}_{=0}\right].$$

Given that $\alpha<1-\lambda<2\alpha$, we can show that the equilibrium prices always decrease in λ:

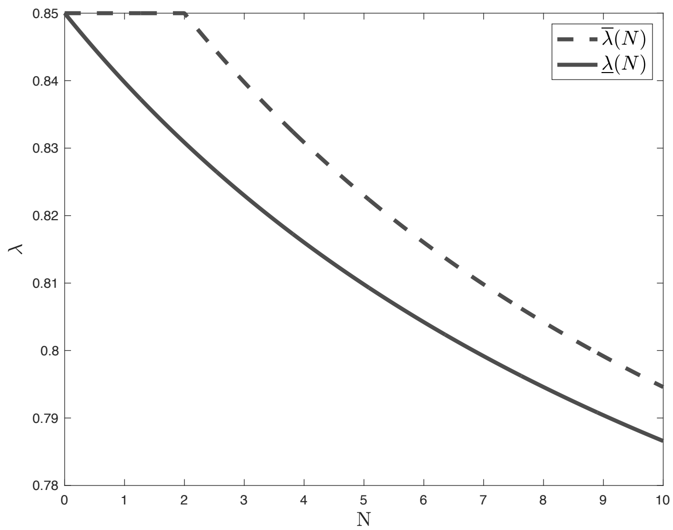
$$\frac{\partial p^*\left(n\right)}{\partial\lambda}=\begin{cases}\left(\frac{2\alpha-(1-\lambda)-n}{4n}\right)N<0 & \text{if } n\in\left(2\alpha-(1-\lambda),1-\lambda\right]\\ \left(\frac{2\alpha+1-\lambda-3n}{4n}\right)N<\frac{2N(\alpha-(1-\lambda))}{4n}<0 & \text{if } n\in(1-\lambda,n_2)\end{cases}.$$

Therefore we must have $\frac{\partial\Pi}{\partial\lambda}<0$.

Note that when $\alpha\in(0,\frac{1}{6})$, $M(\cdot)$ in cases (ii) and (iii) correspond to regions 1,2,5 if $\lambda\in(1-2\alpha,\frac{1}{2}+2\alpha)$ and regions 1,2,5,6 if $\lambda\in(\frac{1}{2}+2\alpha,1-\alpha)$. However, the addition of region 6 does not change the boundary conditions as $M(\cdot)$ strictly declines in n in region 6. To summarize, we let $\underline{\lambda}\equiv1-\frac{\left(N+4+2\sqrt{2}\right)\alpha}{\frac{N}{2}+4+2\sqrt{2}}$ and $\overline{\lambda}\equiv\min\left\{1-\left(\frac{2+2\sqrt{2}+N}{3+2\sqrt{2}+\frac{N}{2}}\right)\alpha,1-\alpha\right\}$. Hence we have $\frac{\partial\Pi}{\partial\lambda}<0$ if $\lambda\in\left(\underline{\lambda},\overline{\lambda}\right)$ and $\overline{\lambda}\equiv\frac{\partial\Pi}{\partial\lambda}\geq0$ if $\lambda\in\left(\overline{\lambda},1\right)$. Figure A1 illustrates how $\underline{\lambda}$ and $\overline{\lambda}$ changes with respect to N. Specifically, $\overline{\lambda}$ converges to $\underline{\lambda}$ when N approaches 0 or infinity.

□

Figure A1: $\overline{\lambda}(N)$ and $\underline{\lambda}(N)$



Note: $\alpha = 0.15$.

A.2 Additional Analyses

A.2.1 Platform's Filtering Strategy: Optimal Threshold n_0

One potential way for the platform to neutralize the negative impact of matching accuracy improvement is recommending only a subset of AI-selected influencers to marketers by filtering out niche influencers with n less than some threshold n_0. The platform needs to optimally choose the threshold $n_0 \in [\underline{n}, \frac{1}{2}]$ according to the matching accuracy λ (as well as the follower density ρ) to potentially benefit from this filtering action.

When the threshold is n_0, influencers with $n \in [n_0, \frac{1}{2}]$ participate. Clearly, the indifference condition still applies, so $M(n) - p(n)$ is a constant. For the marginal marketer $\bar{c}$, we still have $M(n) - p(n) - \bar{c} = 0$. In addition, market clearing implies $\frac{1}{2} - n_0 = \bar{c}$. Thus, we have $p(n) = M(n) - \frac{1}{2} + n_0$. The platform chooses n_0 to maximize its revenue given by

$$\Pi(n_0) = \delta \int_{n_0}^{\frac{1}{2}} p(n)dn = \delta \int_{n_0}^{\frac{1}{2}} [M(n) - \frac{1}{2} + n_0]dn. \tag{A29}$$

The first-order condition implies

$$\Pi'(n_0) = \delta(1 - 2n_0 - M(n_0)) = 0, \tag{A30}$$

or

$$1 - 2n_0 - M(n_0) = 0. \tag{A31}$$

Note that $\Pi'(n_0)$ is clearly decreasing in n_0, with $\Pi'(\frac{1}{2}) = -M(\frac{1}{2}) < 0$ and $\Pi'(\underline{n}) = 1 - 2\underline{n} - M(\underline{n}) >$

$\frac{1}{2}-\underline{n}-M(\underline{n}) = p^*(\underline{n}) = 0$, where the inequality comes from the fact that $\underline{n} < \frac{1}{2}$ and the last equality is supported by Lemma 2. Therefore, $1 - 2n_0 - M(n_0) = 0$ has a unique solution $n_0 \in [\underline{n}, \frac{1}{2}]$.

When n_0 is optimally chosen by the platform, its revenue never decreases as matching accuracy λ increases. To see this, consider the following thought experiment. When the matching accuracy increases from λ_1 (with optimal $n_0|_{\lambda=\lambda_1}$) to $\lambda_2 > \lambda_1$, the platform can at least maintain this threshold and obtain commission revenue from the same set of influencers who now charge higher equilibrium prices. Thus, the platform's revenue does not decrease in λ, i.e., $\Pi|_{\lambda=\lambda_2} \geq \delta \int_{n_0|_{\lambda=\lambda_1}}^{\frac{1}{2}} [M(n)|_{\lambda=\lambda_2} - \frac{1}{2} + n_0|_{\lambda=\lambda_1}]dn \geq \delta \int_{n_0|_{\lambda=\lambda_1}}^{\frac{1}{2}} [M(n)|_{\lambda=\lambda_1} - \frac{1}{2} + n_0|_{\lambda=\lambda_1}]dn = \Pi|_{\lambda=\lambda_1}$.

A.2.2 Heterogeneity in Follower Size vs. Content Broadness: Both ρ and n Are Random Variables

Here, we consider a comprehensive model where both the follower density $\rho \in [0, \rho_0]$ and content broadness $n \in [0, \frac{1}{2}]$ of influencers are random variables and they are allowed to be either positively or negatively correlated. To model the dependency structure between these two variables, we construct a bivariate Frank copula (Schmidt, 2007) ($u = \frac{\rho}{\rho_0} \in [0, 1]$, $v = 2n \in [0, 1]$) where the marginal distribution of u and v are both uniform in $[0, 1]$ and the joint cumulative distribution function of (u, v) is given by

$$C_\theta(u, v) = \frac{1}{\theta}\log\left[1 + \frac{\left(e^{-\theta u} - 1\right)\left(e^{-\theta v} - 1\right)}{e^{-\theta} - 1}\right], \theta \in (-\infty, +\infty) \setminus \{0\}.$$

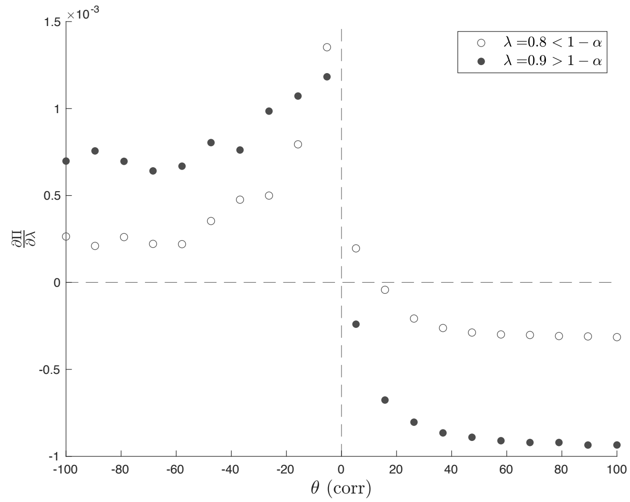
When $\theta > 0$ ($\theta < 0$), ρ and n are positively (negatively) correlated and the absolute value of θ indicates the strength of this dependency. With the added dependency structure, this model is analytically hard to solve so we numerically explore the relationship between the point derivative of profit w.r.t. λ and the correlation coefficient θ at given values of λ. From the analysis of our baseline model and model (1) in Section 4.5, we conjecture that at a given λ, the platform revenue is less (more) likely to decline with AI improvement when ρ and n are negatively (positively) correlated, which is confirmed by Figure A2.

A.2.3 Alternative Model for Matching Accuracy

In the baseline model, we model the inaccuracy of AI as a fixed distance ($\frac{\kappa}{2}$) between the target consumers and the core followers of influencers. This modeling choice is mathematically convenient, but it implies that all influencers who are competing with each other have the same core followers ($\hat{x} = \frac{\kappa}{2}$). In this extension, we relax this constraint by proposing an alternative way of modeling matching accuracy. We show that there is no qualitative change in our results.

In the alternative model, we model the inaccuracy of AI as follows. The platform's matching algorithm recommends to a brand with target consumers at $x = 0$ a group of influencers with $n \in$

Figure A2: The point derivative of profit ($\frac{\partial \Pi}{\partial \lambda}$) vs. correlation coefficient (θ)



Note: $\alpha = 0.15$, $\rho_0 = 10$, and $\lambda = 0.8, 0.9$.

$[0, \frac{1}{2}]$, where the core followers of each influencer is randomly sampled between $[-\frac{\epsilon}{2}, \frac{\epsilon}{2}]$ $(0 \leq \epsilon \leq \frac{1}{2})$. That is, unlike the baseline model where κ is a fixed number, here we assume that

$$\kappa \sim U[-\epsilon, \epsilon], \tag{A32}$$

so the core followers of influencers are located at $\frac{\kappa}{2} \sim U[-\frac{\epsilon}{2}, \frac{\epsilon}{2}]$.

Note that although *ex post* influencers with different core followers will participate in influencer marketing campaigns and thus their distances to the target consumers will differ, influencers do not know their distance to the marketer that collaborates with them *ex ante* when setting the prices since the matching process is random. That is, an influencer do not have more information about the distance between her and the matched marketer before the match is realized because the prices are posted *ex ante*.

Influencers and marketers make their pricing and participation decisions based on the *ex ante* expected converted revenue of influencer marketing, denoted by $\overline{M}(n)$. Similar to the baseline model, we let $\overline{m}(n) \equiv \overline{M}(n)/\rho$. Since $\overline{m}(n)$ is the *ex ante* expected revenue, it is in fact an average of $m(n)$ in our baseline model[25]:

$$\overline{m}(n) = \mathbf{E}_{\kappa}[m(n)] = \frac{1}{\epsilon} \int_0^{\epsilon} m(n) d\kappa, \tag{A33}$$

[25] By symmetry, it is sufficient to focus on $\kappa > 0$.

and by the proof of Lemma 1, we already know that

$$m(n) = \begin{cases} ① \ n\left(\alpha - \frac{\kappa}{2}\right) & \text{if } n \leq \kappa \text{ and } n \leq 2\alpha - \kappa, \\ ② \ \frac{(n-\kappa+2\alpha)^2}{8} & \text{if } n \leq \kappa \text{ and } n > |2\alpha - \kappa|, \\ ③ \ 0 & \text{if } n \leq \kappa \text{ and } n \leq \kappa - 2\alpha, \\ ④ \ n\alpha - \frac{\kappa^2+n^2}{4} & \text{if } n > \kappa \text{ and } n \leq 2\alpha - \kappa, \\ ⑤ \ \frac{4\alpha^2+4\alpha(n-\kappa)-(n-\kappa)^2}{8} & \text{if } n > \kappa \text{ and } 2\alpha - \kappa < n < 2\alpha + \kappa, \\ ⑥ \ \alpha^2 & \text{if } n > \kappa \text{ and } n \geq 2\alpha + \kappa. \end{cases} \tag{A34}$$

Plugging in the expression of $m(n)$ to equation (A33) and using some algebra to simplify the expression gives

$$\overline{m}(n) = \begin{cases} ① \ \frac{n(3\epsilon(4\alpha-\epsilon)-n^2)}{12\epsilon} & \text{if } n \leq \epsilon \text{ and } n \leq 2\alpha - \epsilon, \\ ② \ \frac{n\alpha^2}{\epsilon} - \frac{(n-\epsilon+2\alpha)^3}{24\epsilon} & \text{if } n \leq \epsilon \text{ and } n > |2\alpha - \epsilon|, \\ ③ \ \frac{n\alpha^2}{\epsilon} & \text{if } n \leq \epsilon \text{ and } n \leq \epsilon - 2\alpha, \\ ④ \ n\alpha - \frac{\epsilon^2/3+n^2}{4} & \text{if } n > \epsilon \text{ and } n \leq 2\alpha - \epsilon, \\ ⑤ \ \alpha^2 - \frac{(\epsilon-n+2\alpha)^3}{24\epsilon} & \text{if } n > \epsilon \text{ and } 2\alpha - \epsilon < n < 2\alpha + \epsilon, \\ ⑥ \ \alpha^2 & \text{if } n > \epsilon \text{ and } n \geq 2\alpha + \epsilon. \end{cases} \tag{A35}$$

One can verify that $\overline{M}(n)$ is increasing in n. The structure of $\overline{m}(n)$ is similar to that of $m(n)$ but mathematically more complicated. Similar to our baseline model, we introduce the parameter for matching accuracy $\hat{\lambda} \equiv 1 - \epsilon$ and focus only on the case when the matching accuracy is large enough: $\hat{\lambda} \in \left[\max\left\{\frac{1}{2} + 2\alpha, 1 - \alpha\right\}, 1\right]$.

The derivation of equilibrium is the same as that for the baseline model. We know that marketers and influencers participate if and only if $c \in \mathcal{C} = [0, \overline{c}]$ and $n \in \mathcal{N} = [\underline{n}, \frac{1}{2}]$ as in Lemma 2, and in equilibrium we have

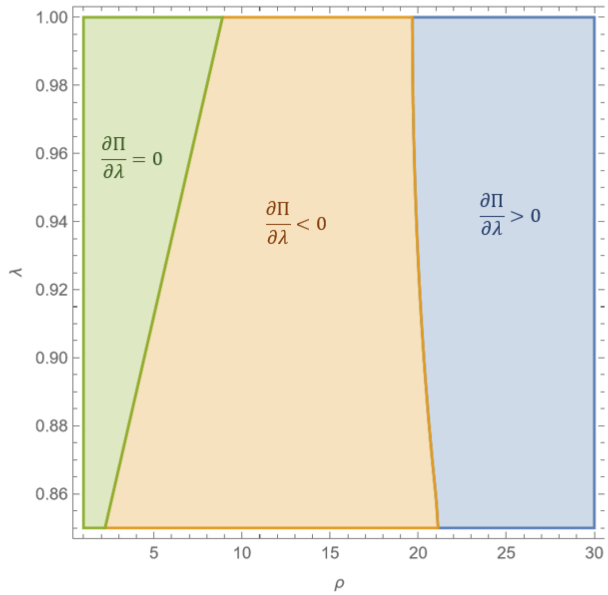
$$\overline{M}(\frac{1}{2} - \overline{c}) = \overline{c}. \tag{A36}$$

Similar to equation (A4), the following equation still holds:

$$\frac{\partial \Pi}{\partial \lambda} = \delta \left[\int_{\underline{n}}^{\frac{1}{2}} \frac{\partial \overline{M}(n)}{\partial \lambda} dn + \frac{\partial \overline{c}}{\partial \lambda}(-\overline{c}) \right]. \tag{A37}$$

However, due to the mathematical complexity of $\overline{M}(n)$ in this alternative model of matching accuracy, we rely on a numerical analysis that calculates $\frac{\partial \Pi}{\partial \lambda}$ to show the robustness of our results in Proposition 1. Figure A3 illustrates the results of the numerical analysis. From Figure A3, we can see that for any λ, $\frac{\partial \Pi}{\partial \lambda} = 0$ when ρ is small, $\frac{\partial \Pi}{\partial \lambda} < 0$ when ρ is intermediate, and $\frac{\partial \Pi}{\partial \lambda} > 0$ when

Figure A3: Sign of $\frac{\partial \Pi}{\partial \lambda}$



Note: $\alpha = 0.15$.

ρ is large. This is consistent with the findings in Proposition 1.

A.2.4 Discussions on Competition: Technical Details

A.2.4.1 Brands Competing for Consumer Demand

In our baseline model, we assume that marketers' revenue earned from a consumer is independent of the level of competition among brands. In this section, we explore the robustness of our results by incorporating the product-level competition among brands for consumer demand into our model. We first propose a model slightly modified from our baseline model to illustrate the main forces behind this competition. We assume that the expected revenue a brand (with target consumers at 0) can obtain from a consumer at x is discounted by $(1 - |\mathcal{C}|)$, where $|\mathcal{C}|$ measures the number of participating marketers who purchase endorsements from influencers.[26] To differentiate from the baseline model, we add a "$\sim$" on top of all key variables. Therefore, the converted revenue for a marketer to collaborate with an influencer with content broadness n becomes:

$$\tilde{M}(n) = (1 - |\mathcal{C}|)M(n). \tag{A38}$$

Similar to the main model, we can show that marketers with low enough cost c ($c \leq \tilde{\bar{c}}$) participate in equilibrium, and thus $|\mathcal{C}| = \tilde{\bar{c}}$. Thus, we have influencers with $n \geq \tilde{\underline{n}}$ participate in the influencer marketing campaigns since $\tilde{M}(n)$ is increasing in n for any given $|\mathcal{C}|$. Note that in this extended

[26]We can also let the discount factor be $(1 - \beta|\mathcal{C}|)$, where $0 \leq \beta \leq 2$. The results do not change qualitatively.

model, equilibrium conditions imply that

$$(1-\tilde{\bar{c}})M(\frac{1}{2}-\tilde{\bar{c}})=\tilde{\bar{c}}, \tag{A39}$$

whereas in the baseline model we have $M(\frac{1}{2}-\bar{c})=\bar{c}$. Given that $M(\cdot)$ is an increasing function, it is easy to see that, when the consumer demand is affected by competition among brands, the mass of participating marketers and influencers is less than that in the baseline model, i.e., $\tilde{\bar{c}}<\bar{c}$.

In our baseline model, we can rewrite equation (12) and decompose the impact of matching accuracy on platform revenue into the following total matching enhancement effect and total competition effect on all participating influencers.

$$\frac{\partial \Pi}{\partial \lambda}=\delta\left[\underbrace{\int_{\frac{1}{2}-\bar{c}}^{\frac{1}{2}}\frac{\partial M(n)}{\partial \lambda}dn}_{\text{total matching enhancement effect}}-\underbrace{\bar{c}\cdot\frac{\partial \bar{c}}{\partial \lambda}}_{\text{total competition effect}}\right]. \tag{A40}$$

In this extension, instead, we have

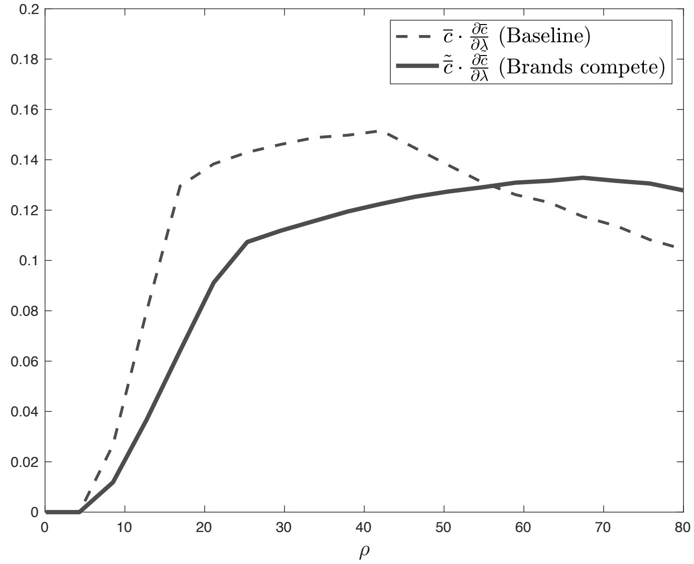
$$\frac{\partial \tilde{\Pi}}{\partial \lambda}=\delta\left[\int_{\frac{1}{2}-\tilde{\bar{c}}}^{\frac{1}{2}}\left(\frac{\partial[(1-\tilde{\bar{c}})M(n)]}{\partial \lambda}-\frac{\partial \tilde{\bar{c}}}{\partial \lambda}\right)dn\right] \tag{A41}$$

$$=\delta\left[\underbrace{\int_{\frac{1}{2}-\tilde{\bar{c}}}^{\frac{1}{2}}\left(\frac{\partial M(n)}{\partial \lambda}(1-\tilde{\bar{c}})-M(n)\frac{\partial \tilde{\bar{c}}}{\partial \lambda}\right)dn}_{\text{total matching enhancement effect}}-\underbrace{\tilde{\bar{c}}\cdot\frac{\partial \tilde{\bar{c}}}{\partial \lambda}}_{\text{total competition effect}}\right]. \tag{A42}$$

Comparing equation (A42) to equation (A40), we can see that the total matching enhancement effect in this extension is "diluted" compared to that in our baseline model. This dilution effect comes from three sources: first, for every participating marketer, the benefit of a higher matching accuracy on its revenue is discounted by its competitors targeting the same consumer segment; second, this discount factor (reflecting the level of competition) itself declines in the number of competitors as improved matching attracts more marketers to participate in equilibrium. Third, due to the discounted revenue from influencer marketing, fewer marketers in total have incentives to collaborate with influencers which leads to fewer participating influencers. On the other hand, the total competition effect may be higher or lower than that in the baseline model as shown in Figure A4. In particular, when ρ is sufficiently small, the marginal influencers are more likely to be general influencers who are affected more by intensified competition due to their diverse follower base, thus $\frac{\partial \tilde{\bar{c}}}{\partial \lambda}<\frac{\partial \bar{c}}{\partial \lambda}$ is more likely to occur.

As a result, we conjecture that $\frac{\partial \tilde{\Pi}}{\partial \lambda}<\frac{\partial \Pi}{\partial \lambda}$ when ρ is large, which is confirmed in Figure A5 generated by a numerical exercise. To summarize, this exercise shows the robustness of our result in

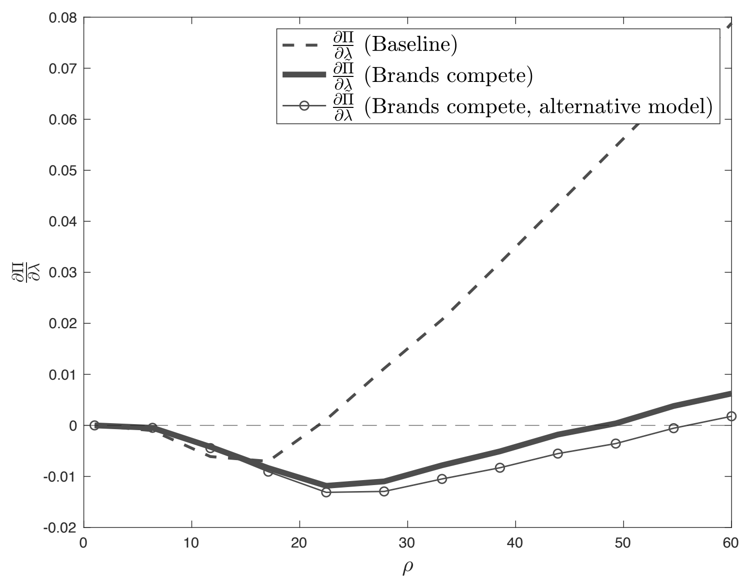
Figure A4: Comparison of Total Competition Effect $\bar{c} \cdot \frac{\partial \bar{c}}{\partial \lambda}$ vs. $\tilde{\bar{c}} \cdot \frac{\partial \tilde{\bar{c}}}{\partial \lambda}$



Note: $\alpha = 0.15$ and $\lambda = 0.9$.

that the shape of $\frac{\partial \tilde{\Pi}}{\partial \lambda}$ with respect to the follower density ρ is similar to that in our baseline model. However, the competition among brands for consumer demand dilutes the positive matching enhancement effect, whereas it could strengthen or weaken the negative competition effect, depending on the level of follower density ρ.

Figure A5: Comparison of Total Effect $\frac{\partial \Pi}{\partial \lambda}$ vs. $\frac{\partial \tilde{\Pi}}{\partial \lambda}$

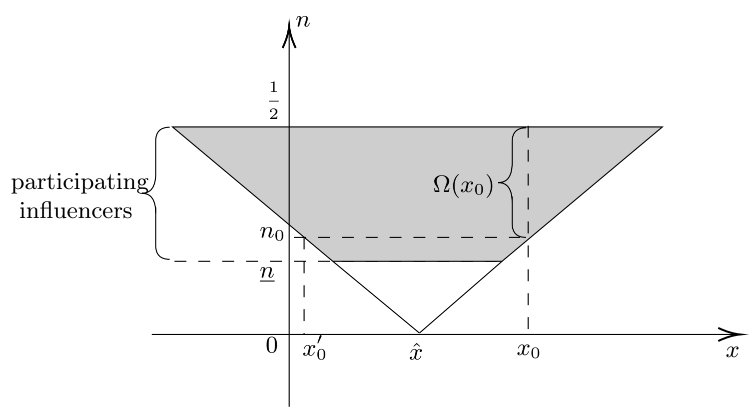


Note: $\alpha = 0.15$ and $\lambda = 0.9$.

In the illustrative model above, we have assumed that the competition-based discount factor is uniform across all consumers. Nevertheless, one may argue that consumers who are closer to the core followers of influencers ($\hat{x}$) should be exposed to more marketing campaigns than those further away from $\hat{x}$, because consumers closer to $\hat{x}$ can be reached by almost all influencers while consumers further away from $\hat{x}$ can only be reached by more general influencers. Next we propose an alternative model that precisely models the varying intensity of influencer marketing that consumers at different locations are exposed to.

Alternative Model for Brand Competition Now we assume the expected revenue a brand (with target consumers at 0) can obtain from a consumer at x is discounted by $(1 - \Omega(x))$, where $\Omega(x)$ measures the number of participating influencers that *reach* x. Figure A6 illustrates this assumption, where the x-axis indicates the location of a consumer (x) and the y-axis indicates the content broadness of influencers (n). For instance, the influencer with n_0 reaches consumers located between x'_0 and x_0. The shaded area in Figure A6 indicates the reach of consumers from all participating influencers in equilibrium. The consumers located at x_0 are exposed to all influencers with $n \in [n_0, \frac{1}{2}]$ who reach them.

Figure A6: Illustration of $\Omega(x)$



According to Figure A6, the expression of $\Omega(x)$ is given by:

$$\Omega(x) = \frac{1}{2} - \max\{2|x - \hat{x}|, \underline{n}\} = \frac{1}{2} - \max\{2|x - \frac{1-\lambda}{2}|, \underline{n}\}, \tag{A43}$$

where λ is the matching accuracy. Thus, a marketer's total expected revenue ($\tilde{M}(n)$) from collaborating with influencer n is

$$\tilde{M}(n) = \rho \int_{\frac{1-\lambda}{2} - \frac{n}{2}}^{\frac{1-\lambda}{2} + \frac{n}{2}} \max\{0, (1 - \Omega(x))(\alpha - |x|)\}\, dx. \tag{A44}$$

Now that the discount factor depends on x, the expression of $\tilde{M}(n)$ is more complicated and this model is analytically much more challenging to solve. Therefore, we provide the results from a numerical simulation in Figure A5 comparing $\frac{\partial \Pi}{\partial \lambda}$ in the baseline model, the first model in this extension, and this alternative model to show the robustness of our results. Compared to the first model above, the dilution effect on matching enhancement is even more salient in this alternative model because it accounts for the fact marketers compete on consumers closer to their target consumers as the matching accuracy increases.

A.2.4.2 One-to-Many Endorsement

In our baseline model, we assume that each influencer can collaborate with at most one marketer. This assumption represents the case where marketers compete with each other so that each influencer can only endorse one brand. In practice, however, some influencers do endorse multiple non-competing brands. Our baseline model can be easily modified to allow for influencers working with up to $k \in \mathbf{N}$ multiple brands. We first prove that in the new equilibrium, all participating influencers work with k marketers because the indifference condition for the marketers are still satisfied. The following proposition shows that allowing influencers to offer their services to multiple brands can lead to lower equilibrium prices due to intensified competition from market expansion.

Proposition A.1. *For any participating influencer $n \geq \underline{n}$, their equilibrium price $p^*(n)$ strictly decreases in the maximum number of brands k that each influencer can endorse.*

Note that this price drop effect is symmetric to all influencers and it is orthogonal to the AI improvement. Hence our analysis remains the same when influencers are allowed to collaborate with more than one marketer, as long as all types of influencers are endowed with equal opportunities to endorse multiple marketers. However, our analysis so far has restricted all marketers (brands) that an influencer collaborates with to have the same target consumers. In practice, general influencers may collaborate with more marketers with different target consumers (e.g., marketers from different product categories) than the niches ones due to their diverse follower base. When general influencers can collaborate with more marketers than niche ones, this competition effect does interact with AI improvement because of its asymmetric impact on different types of influencers. It is analytically challenging to extend our framework to allow influencers to promote multiple marketers with different target consumers. Hence, we present a simple model to discuss the main insights.

Simple Model: Influencers Promoting Marketers with Different Target Consumers Suppose there are two marketers A and B with different target consumers. There are two niche influencers n_A and n_B and one general influencer n_G. Suppose n_A is suitable for marketer A and n_B is suitable for marketer B. On the one hand, a marketer obtains a revenue of M_N only if it works with the "suitable" niche influencer and it obtains zero revenue if working with the unsuitable niche influencer. On the other hand, both marketers obtain a revenue of M_G if they work with the general influencer who is suitable for both marketers. The two marketers A and B incur a respective cost of c_A and c_B from working with influencers, where $0 < c_A < c_B < M_G < M_N$. To highlight the basic forces, we further assume that $M_G < 2c_B - c_A$ so the general influencer would prefer to work with only marketer A at a high price when there is no competition from the niche influencers. AI recommends $\{n_A, n_G\}$ to marketer A with probability $\tau \in [0, 1]$ and to marketer B with probability $1-\tau$. Likewise, it recommends $\{n_B, n_G\}$ to marketer B with probability τ and to marketer A with probability $1 - \tau$. Here, τ measures thematching accuracy, and the expected revenue from working with a niche influencer is $\tau \cdot M_N$ for both marketers. Consider the following five cases:

Case (i) $\tau \cdot M_N < c_A$.

Neither marketer would ever work with niche influencers even at zero prices. The general influencer n_G works with only marketer A at a price of $p^*(n_G) = M_G - c_A$.

Case (ii) $c_A \le \tau \cdot M_N < 2c_B - M_G$.

Now that the expected revenue from working with a niche influencer is higher than marketer A's cost, in order to compete with the niche influencers, the general influencer can no longer charge a price higher than $M_G - \tau \cdot M_N$. But the general influencer continues to work with only marketer A, albeit at a lower price of $p^*(n_G) = M_G - \tau \cdot M_N$.

Case (iii) $2c_B - M_G \le \tau \cdot M_N < c_B$.

In this case, the expected revenue from working with a niche influencer is sufficiently high so it imposes such a downward pressure on the pricing power of the general influencer that the general influencer might as well collaborate with both marketers by further lowering her price to $p^*(n_G) = M_G - c_B$.

Case (iv) $c_B \le \tau \cdot M_N < M_G$.

In this case, the general influencer continues to work with both marketers at an even lower price $p^*(n_G) = M_G - \tau \cdot M_N$, because of the competitive pressure from niche influencers in both product categories.

Case (v) $\tau \cdot M_N > M_G$.

Now that the niche influencers become more attractive to marketers due to sufficiently good matching technology, both marketers prefer to work with only niche influencers at a price of $p^*(n_N) = \tau \cdot M_N - M_G$.

In sum, the platform's revenue Π is given by

$$\Pi = \begin{cases} \delta\left(M_G - c_A\right) & \text{if } \tau < \frac{c_A}{M_N} \\ \delta\left(M_G - \tau \cdot M_N\right) & \text{if } \frac{c_A}{M_N} \le \tau < \frac{2c_B - M_G}{M_N} \\ 2\delta\left(M_G - c_B\right) & \text{if } \frac{2c_B - M_G}{M_N} \le \tau < \frac{c_B}{M_N} \\ 2\delta\left(M_G - \tau \cdot M_N\right) & \text{if } \frac{c_B}{M_N} \le \tau < \frac{M_G}{M_N} \\ 2\delta\left(\tau \cdot M_N - M_G\right) & \text{if } \tau \ge \frac{M_G}{M_N} \end{cases}.$$

In this simple model, the platform's revenue first decreases and then increases in the matching accuracy τ. When the matching accuracy increases, the general influencer lowers her equilibrium price due to not only the competitive pressure from more suitable niche influencers within the product category, but also their incentive of switching to serving more product categories as they face intensified competition from one product category. In other words, enhanced matching leads to higher within-product-category competition among influencers and it spills over to competition in other product categories as well, exacerbating the potential decline in platform revenue.

Proof of Proposition A.1. We prove by contradiction that in the new equilibrium, all participating influencers work with k marketers. First, we show that the indifference condition still holds in an equilibrium. Suppose there exist two influencers n_1 and n_2 of different sizes where n_1 is matched to marketer c_1 and n_2 to marketer c_2 in equilibrium. Meanwhile, suppose that marketer c_2 strictly prefers to collaborate with n_2 than with n_1, that is,

$$U(n_2, c_2, p^*(n_2)) > U(n_1, c_2, p^*(n_1)).$$

Then by equation (6), we have

$$M(n_2) - p^*(n_2) - c_2 > M(n_1) - p^*(n_1) - c_2,$$

which implies

$$M(n_2) - p^*(n_2) > M(n_1) - p^*(n_1),$$

and thus

$$U(n_2, c_1, p^*(n_2)) = M(n_2) - p^*(n_2) - c_1 > M(n_1) - p^*(n_1) - c_1 = U(n_1, c_1, p^*(n_1)).$$

By proposing a slightly higher-than-equilibrium price $p'(n_2) > p^*(n_2)$ such that $U(n_2, c_1, p'(n_2)) > U(n_1, c_1, p^*(n_1))$ still holds, influencer n_2 would still have someone to collaborate with (marketer c_1) and become strictly better off since the price is higher. This contradicts the notion of an equilibrium.

Next, suppose there exists an equilibrium where an influencer n collaborates with $1 \leq l(< k)$ marketers with $c \in \mathcal{C}$ at price $p^*(n)$. The indifference condition implies that another participating marketer $c' < c$ that is currently matched to influencer n' would obtain the same revenue from working with influencer n at price $p^*(n)$. By proposing a slightly-lower-than-equilibrium price $p'(n) = p^*(n) - \varepsilon$ such that $(l + 1) \cdot p'(n) > l \cdot p^*(n)$, all l marketers that are currently matched to influencer n plus marketer c' would strictly prefer to collaborate with n and influencer n would obtain a strictly higher utility as a result. This contradicts the notion of an equilibrium.

Now that each participating influencer collaborates with k marketers, the market clearing condition becomes

$$(\frac{1}{2} - \underline{n}) \cdot k = \overline{c}. \tag{A45}$$

Recall that the marginal participating marketer ($c = \overline{c}$) obtains zero profit from matching with any participating influencers at their proposed prices, and the marginal influencer's ($n = \underline{n}$) equilibrium price is equal to her "effective" opportunity cost. That is, we have $M(\underline{n}) - \overline{c} = \frac{\gamma_0}{1-\delta}$. Substituting out $\underline{n}$ based on equation (A45), we have

$$M(\frac{1}{2} - \frac{\overline{c}}{k}) - \overline{c} = \frac{\gamma_0}{1-\delta}. \tag{A46}$$

Taking derivatives of both sides w.r.t. k we get

$$\frac{\partial M\left(\frac{1}{2} - \frac{\overline{c}}{k}\right)}{\partial n} \cdot \left(\frac{\overline{c}}{k^2} - \frac{\partial \overline{c}}{\partial k} \cdot k\right) - \frac{\partial \overline{c}}{\partial k} = 0. \tag{A47}$$

We can thus rewrite $\frac{\partial \overline{c}}{\partial k}$ as follows:

$$\frac{\partial \overline{c}}{\partial k} = \frac{\frac{\overline{c}}{k^2} \cdot \frac{\partial M(\underline{n})}{\partial n}}{\frac{\partial M(\underline{n})}{\partial n} \cdot k + 1} > 0. \tag{A48}$$

Given the indifference condition, the equilibrium price of any participating influencer $n \geq \underline{n}$ is given by $p^*(n) = M(n) - \overline{c}$. Therefore, we must have $\frac{\partial p^*(n)}{\partial k} = -\frac{\partial \overline{c}}{\partial k} < 0$.

□